DEPARTMENT OF MECHANICAL ENGINEERING

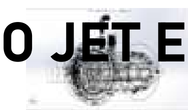
INDIAN INSTITUTE OF TECHNOLOGY ROPAR

RUPNAGAR-140001, INDIA

IIT Ropar

# CP 302
# CAPSTONE PROJECT
## END SEMESTER REPORT

# MICRO JET ENGINE



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# Contents

# 1. ACKNOWLEDGEMENT

We would like to express our deepest gratitude to our esteemed supervisors, **Dr. Rajendra Munian** and **Dr. Sreekanth Shekar Padhee**, for their invaluable guidance, continuous support, and insightful suggestions throughout this project on Micro Jet Engine for UAV Applications. Their expertise and encouragement have played a crucial role in shaping our understanding and approach toward this research.

We also extend our sincere appreciation to the faculty and staff of the **Mechanical Engineering Department, IIT Ropar**, for providing the necessary resources and a conducive environment for our work. Their assistance has been instrumental in the successful completion of this project.

We would also like to acknowledge the various **research papers, technical resources, and prior works** in the field of micro jet engines, which have significantly contributed to our understanding of the subject. The knowledge gained from these references has greatly aided in developing the framework of our study.

A special thanks to our **peers** for their constructive discussions and collaborative spirit, which have enriched our learning experience. Their valuable input and critical feedback have helped refine our ideas and improve our work.

Lastly, we would like to extend our heartfelt gratitude to our **families** for their unwavering encouragement, patience, and motivation. Their constant support has been a source of strength throughout this journey.

# 2. Abstract

This report presents the design, analysis, fabrication progress and control development of a turbine-less microjet engine. Beginning with a compact cycle analysis, we establish the thermodynamic scope (intake → compression → combustion → nozzle expansion) and the equations used to compute state variables, choking conditions and thrust for a range of mass-flow and pressure-ratio scenarios. Using those relations, we summarized the geometric dimensions for key components like diffuser, inner liner, outer liner primary/secondary/dilution zone lengths, hole distributions and nozzle throat/exit areas whose calculations were carried out last semester.

During testing we encountered several practical problems that deviated the real system from ideal cycle predictions like, excessive impeller–diffuser clearance, cantilever-like bending of the compressor assembly causing strong vibrations, low shaft torque and limited RPM (no turbine to drive the shaft), and consequently a reduced compressor pressure ratio and mass flow. To quantify these effects, we generated performance graphs (mass-flow vs RPM, pressure-ratio vs mass-flow, thrust maps), carried out CFD/bench simulations and used venturi/pressure-measurement experiments to obtain accurate pressures. We analytical heat-transfer models for liner wall temperature and conduction were developed to ensure to truly push the engine to its limits.

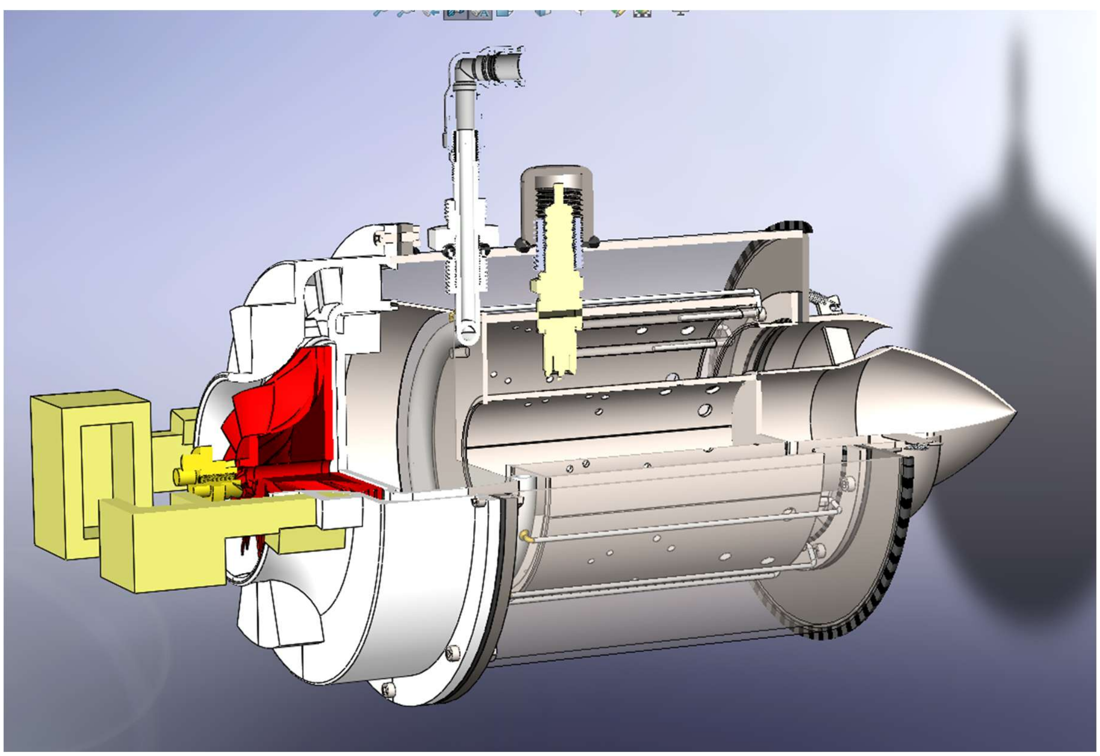


Figure 1 Quarter Sectional View of Microjet Engine

The report also documents our nozzle design approach, fabrication steps (sheet-metal forming, welding, brazing, machining), various challenges faced in fabrication and ESP32-based control/dashboard integration for remote control and data acquisition. Finally, we outline mitigation plans like use of bearings, clearance optimization, improved rotor balancing, and staged compressor redesign and a roadmap to incorporate turbine stages or more robust electric drive solutions to reach the predicted cycle performance by upcoming semester.

# 3. Introduction:

This project focuses on developing a 300 N micro-jet engine for UAV applications, a continuation of the work started last semester. The engine operates on the Brayton cycle, where air is first compressed approximately iso entropically to increase pressure and density, raising its energy potential. Heat is then added at constant pressure in the combustor, increasing the stagnation enthalpy (h). This energy is partly extracted by the turbine to run the compressor, and the remaining enthalpy is converted into jet velocity through isentropic expansion in the nozzle to produce thrust. A higher turbine/nozzle inlet enthalpy corresponds to greater available work, but the maximum temperature is limited by material strength and thermal constraints of turbine blades and combustor walls.

Inside the combustor, a primary zone maintains stable high-temperature combustion, while a dilution zone mixes excess air to cool the gases before they enter turbine components. Not all added heat can be converted into useful work, and some energy is lost to the surroundings, which is why thermal efficiency becomes a key measure of performance.

1. Our goal is to design and build a turbojet engine capable of self-sustaining combustion and delivering 300 N thrust with the highest achievable efficiency. This report summarizes the cycle-analysis basis, component dimensions, practical limitations such as low-pressure ratio, limited RPM, cantilever-type compressor vibration, clearance issues, and the corresponding simulations, heat-transfer analysis, nozzle design approach, fabrication progress, and planned improvements.

# 4. Background

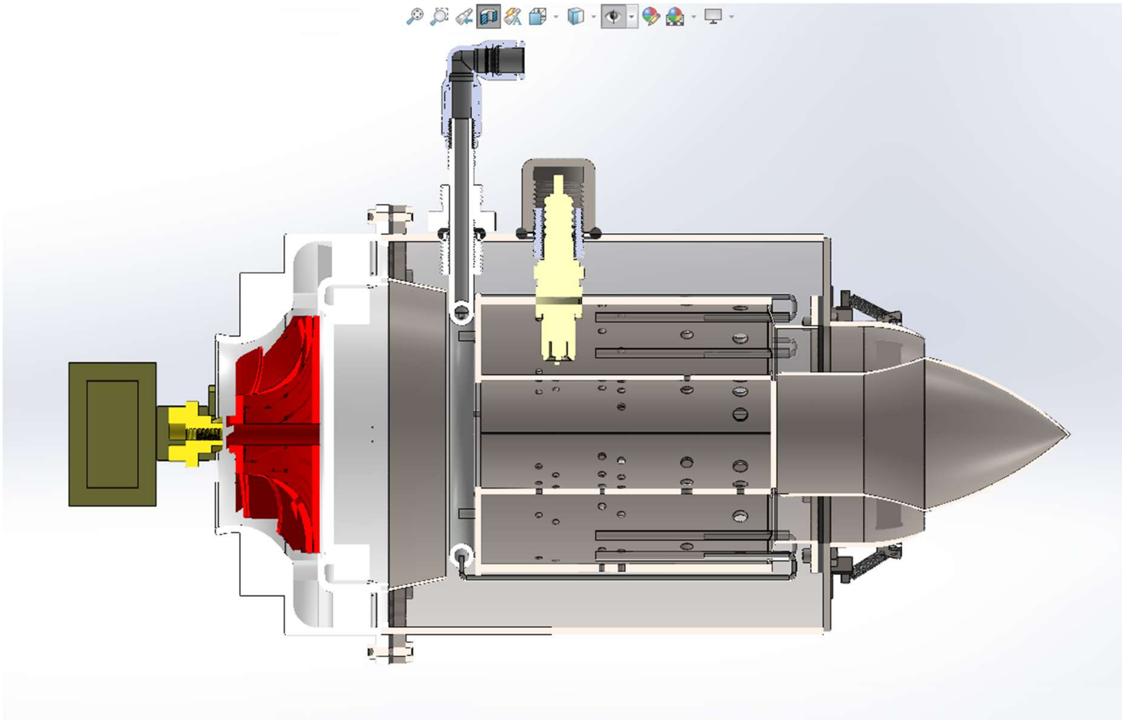


Figure 2 2 Half Sectional View of Micro Jet Engine

## 4.1 CYCLE ANALYSIS

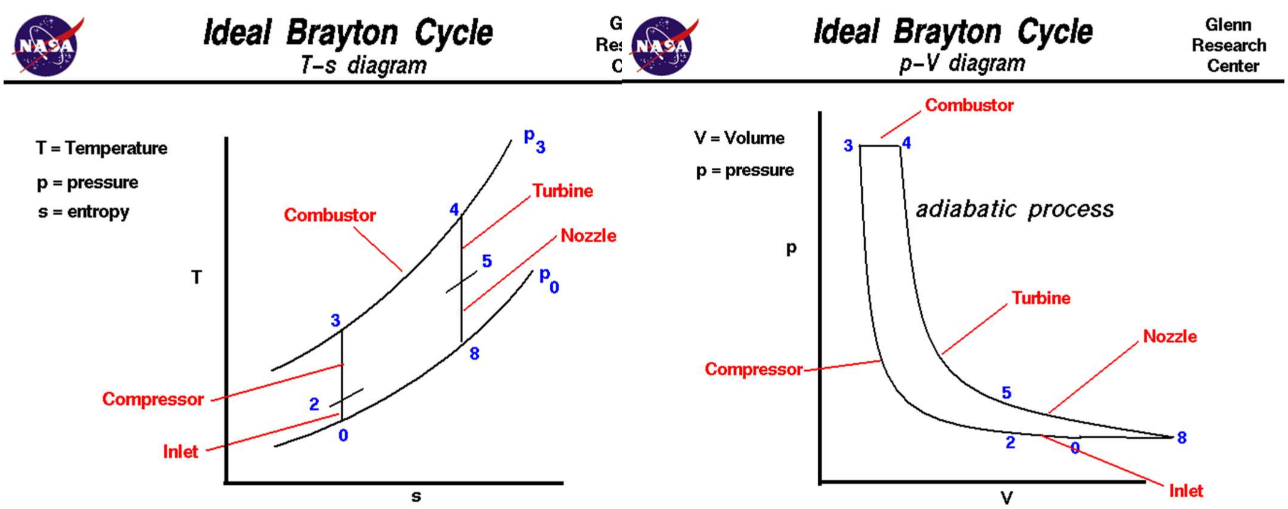



Figure 3 TS & PV diagrams of a Brayton

# 4.2 COMPRESSER:

## Calculations for the state variables:

Nomenclature: We are dealing with compressible flow. So, there will be four variables at each stage i.e., Pressure, Temperature, Density, Velocity flow.

In an incompressible flow, all the state variables and even gas properties are all interdependent on each other making the perfect calculation extremely difficult. So, between the ends of compressor, all gas properties are assumed to be constant as temperature doesn't change that much.

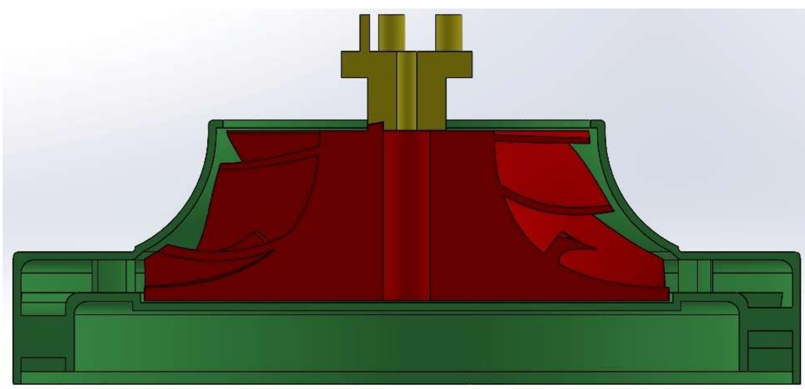
For calculating the non-linear equations, Newton Raphson iteration method is being used.

Once we plot the performance curve of the compressor for the already designed combustion chamber as a partially closed valve, we will directly use the mass flow rate and pressure ratio relation obtained from curve fitting of those plots.

We also measure the pressure P2i using the water column and use that in the calculations done above.

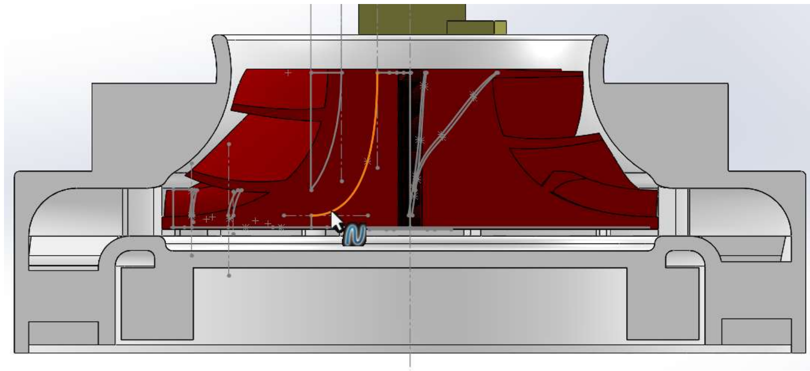
## Design and Making of Centrifugal Compressor:

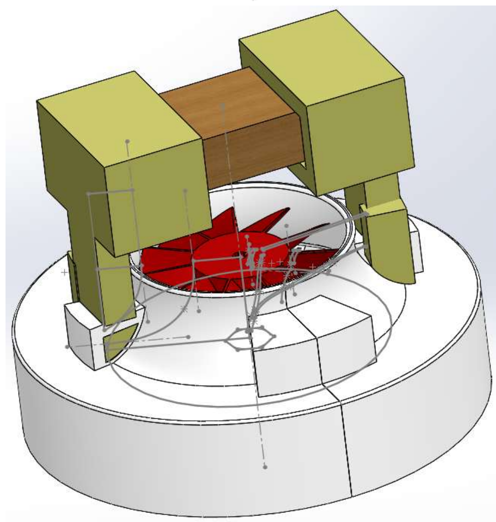
### Previous Design:



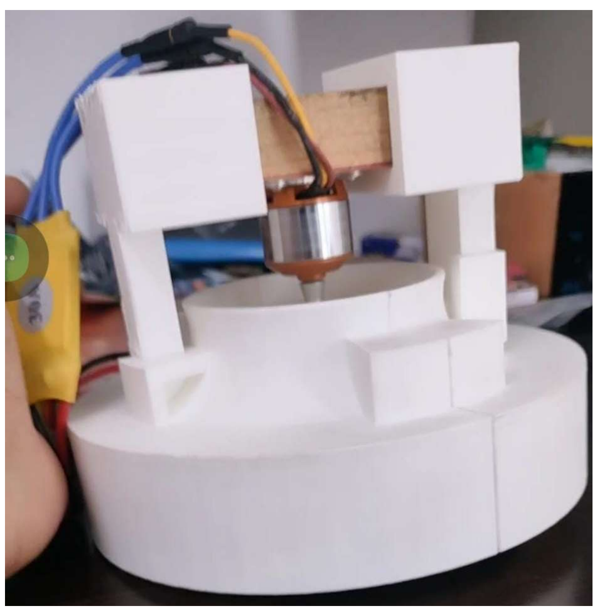
### Modified design:

This completely disassemble design allows us to modify the parts very easily.

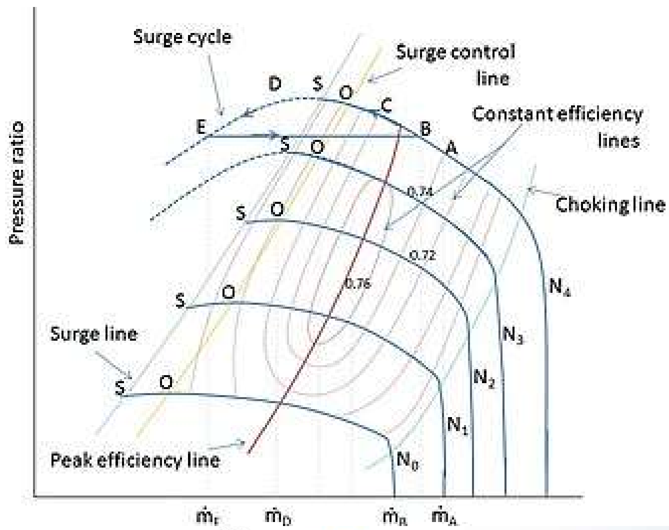




**After Mounting the Motor:**

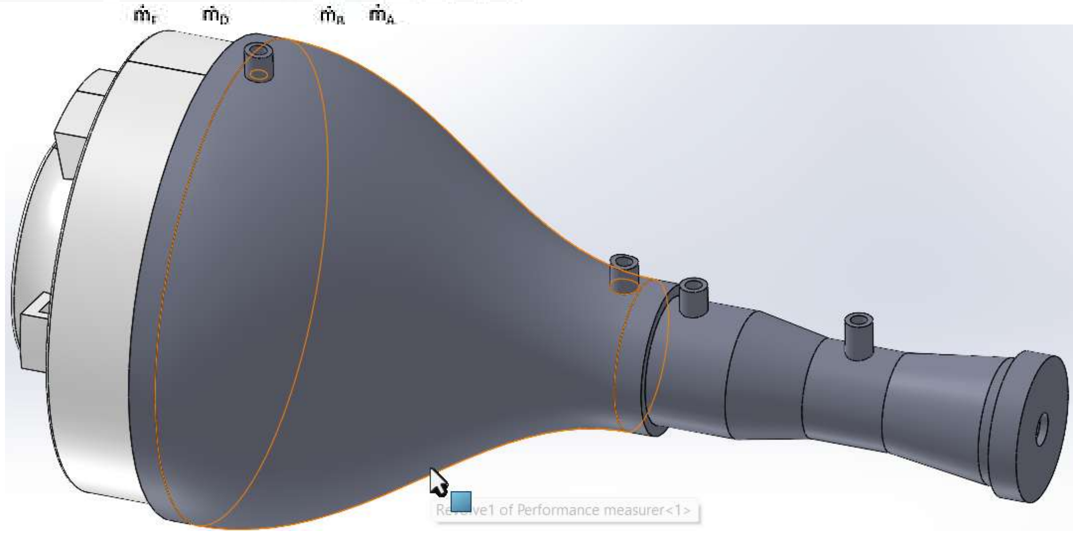


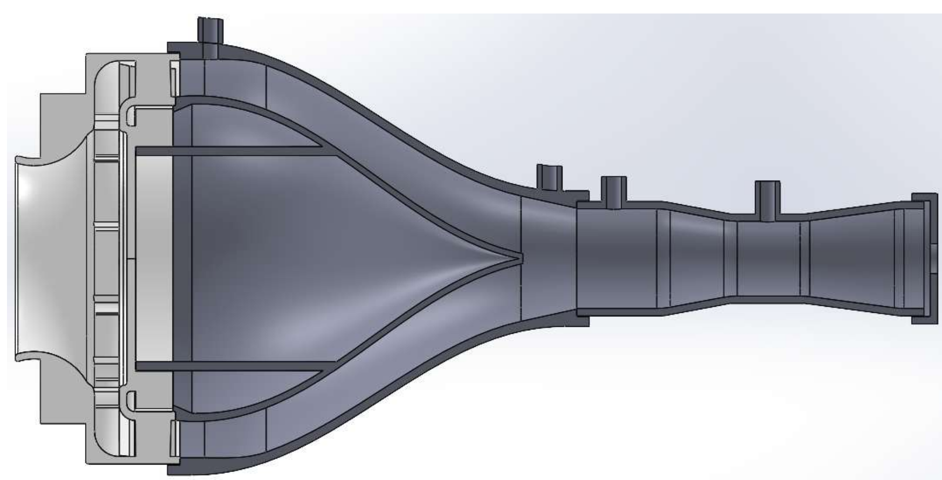
Out next job is to plot the performance curve of the compressor for at least three different RPMs.




We have to find the surge line and avoid the mass flow rate go below the surge line.

- ➢ For finding the performance characteristics, we need to measure two things i.e., pressure at the exit of compressor and the mass flow rate.
- ➢ To measure that, we have designed an Apparatus separately:



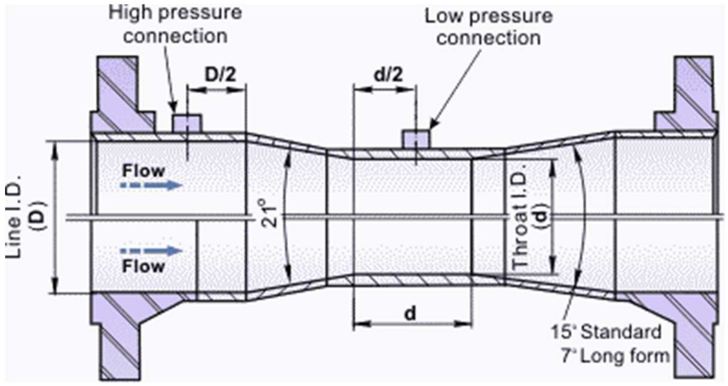


The air flows through the large duct and the venturi meter attached to it at the right end. The first two tube extensions were to measure the static pressure at the two locations. And the remaining two on the venturi meter are to measure the mass flow rate.

Two water level tubes are attached to the larger duct to measure the static pressure and a U-shaped tube goes through the two extensions from venturi meter to measure the pressure difference between the two different cross sections.

## Design of the setup:

Standard design of a venturi meter:




## Mass flow rate expression:

## Mass Flow Rate through an Orifice/Nozzle

$$\dot{m} = C \cdot \varepsilon \cdot A_2 \cdot \sqrt{\frac{2\rho_1 \Delta p}{1-\beta^4}}$$

**Where:**

- **C**: Discharge coefficient (typically 0.985)
- **ε**: Expansibility factor (accounts for gas compressibility)
- **$A_2$**: Area at throat,

$$A_2 = \frac{\pi d^2}{4} \quad \text{where } d \text{ is throat diameter}$$

- **$\rho_1$**: Upstream gas density
- **$\Delta p = p_1 - p_2$**: Differential pressure (inlet – throat)
- **β = d/D :** Diameter ratio (throat diameter / pipe diameter)

## Expansibility Factor (ε)

This corrects for density change as the gas expands passing through the throat. For an ideal gas under isentropic expansion:

$$\varepsilon = \left[\frac{1-\tau^{1/\kappa}}{(1-\beta^4)(1-\tau)}\right]^{1/2}$$

**Where:**

- $\tau = \frac{p_2}{p_1}$: Pressure ratio (throat / inlet)
- $\kappa = \frac{C_p}{C_v}$: Isentropic exponent (ratio of specific heats)
- $\beta = \frac{d}{D}$: Diameter ratio (as defined above)

The design can be at three levels:

1) Design for turbine less engine with constant thrust/RPM:
When there is no turbine, we don't have to start the engine from a minimum thrust value. We can just drive the motor at max RPM and if we just make sure that the combustion chamber's resistance is not letting the compressor into surge, that's enough.

2) Design for turbine less engine with variable thrust/RPM.
For this design, we need to make sure that for a wide range of RPMs, the compressor should not go into surge. For this design, we have to vary the RPM of motor from 100% to the minimum at each of the RPMs, we will plot the mass flow rate vs RPM and Pressure ratio vs RPM.

The mass flow rate and peak pressure data can help us later when examining (Testing) the combustion chamber as the velocity of the air cannot go beyond the optimal range for sustaining the flame.

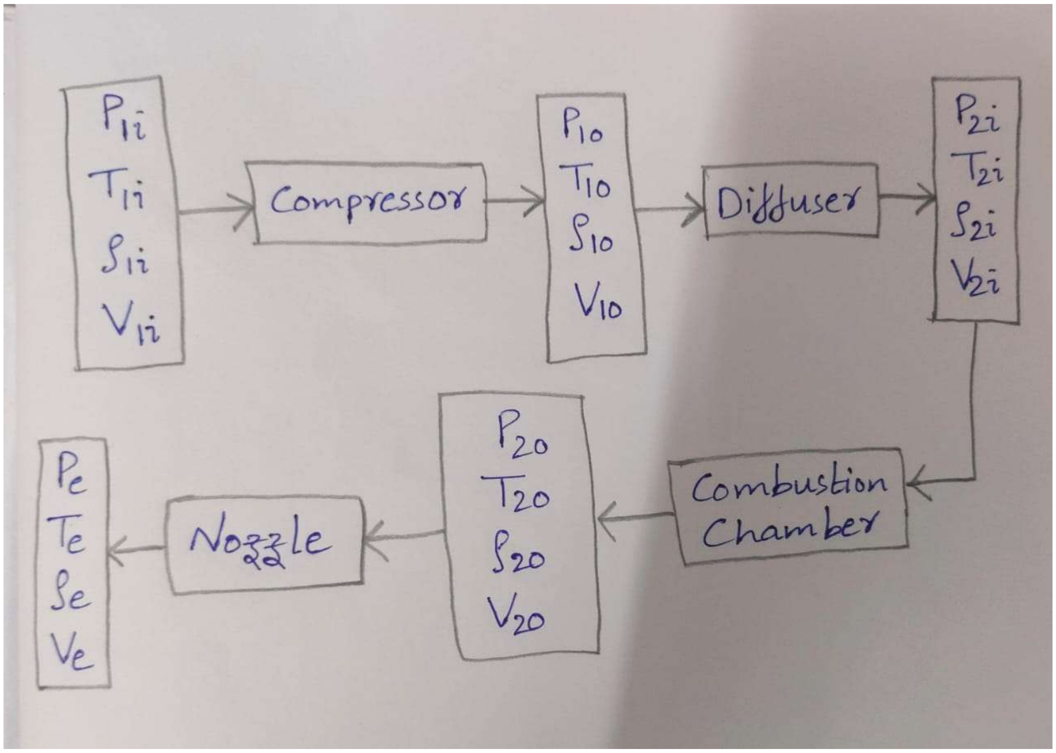
3) Design for engine with turbine with variable Thrust and RPM (Final Goal).
General practice to vary the mass flow rate is to use a flow control valve. Basically, the combustion chamber will act like a partially closed valve due to its intricate flow passages. And we should make sure the combustion chamber as a partially closed calve is not too closed such that the compressor is always above the surge line. And also, as close to the peak pressure point as close as possible at least for a good range of RPMs.

## Calculations for the state variables:

- Nomenclature: We are dealing with compressible flow. So, there will be four variables at each stage i.e., Pressure, Temperature, Density, Velocity flow.
- In a incompressible flow, all the state variables and even gas properties are all interdependent on each other making the perfect calculation extremely

difficult. So, between the ends of compressor, all gas properties are assumed to be constant as temperature doesn’t change that much.

- For calculating the non-linear equations, Newton Raphson iteration method is being used.



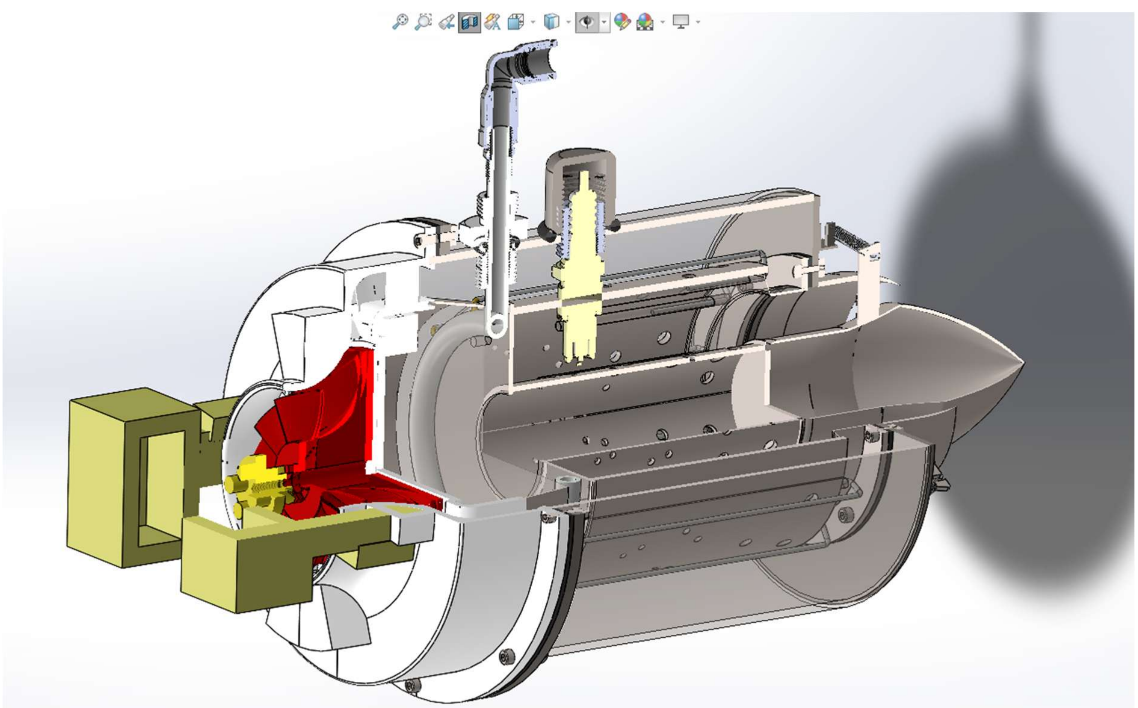



*Figure 4 Quater Sectional View showing compressor ,combustor and Nozzle*

$$\boxed{\begin{aligned} P_{1i} &= 1.01\times10^{5}\,Pa \\ T_{1i} &= 300K \\ \rho_{1i} &= 1.204\,kg/m^3 \\ V_{1i} &= 0\,m/s \end{aligned}} \longrightarrow \text{Before entering the Compressor.}$$

At 10,000 rpm, $\boxed{\dot{m} = 0.0137\,kg/s}$

$$\frac{P_{10}}{P_{1i}} = 1.0132$$

$$\Rightarrow P_{10} = 1.01\times1.0132\times10^{5}\,Pa$$

k

$$\boxed{P_{10} = 1.0233\times10^{5}\,Pa}$$

$$V_{10} = \frac{\dot{m}}{\rho_{10}A_1}$$

$R = 287\,J/kgK$

$$\rho_{10} = f(P_{10}, T_{10}) \longrightarrow \rho_{10} = \frac{P_{10}}{RT_{10}}$$

$$T_{10} = T_{1i}\left[1+\frac{1}{\eta}\left(\left(\frac{P_{10}}{P_{1i}}\right)^{\frac{\gamma-1}{\gamma}}-1\right)\right]$$

If $\eta = 0.85$, $\boxed{T_{10} = 301.32K,\ \rho_{10} = 1.1833\,kg/m^3}$

$$V_{10} = \frac{0.0137}{1.1833\times0.002049} \Rightarrow \boxed{V_{10} = 5.65\,m/s}$$

$$\boxed{\begin{aligned} P_{10} &= 1.0233\times10^{5}\,Pa \\ T_{10} &= 301.32K \\ \rho_{10} &= 1.1833\,kg/m^3 \\ V_{10} &= 5.65\,m/s \\ \dot{m} &= 0.0137\,kg/s \end{aligned}} \longrightarrow \text{Just after exiting the Compressor vanes. (Before entering the diffuser.}$$

Figure 5 Figure State Variable Calculation at Compressor Outlet

$$\dot{m} = \rho_{2i} A_2 V_{2i}$$

$$\rho_{2i} = \frac{P_{2i}}{R T_{2i}}$$

Stagnation temperature should be constant for adiabatic process:

$$T_0 = T + \frac{V_{2i}^2}{2C_p} = \text{Constant}$$

$$301.32 + \frac{(5.65)^2}{2(1.005)} = T_{2i} + \frac{V_{2i}^2}{2(1.005)}$$

$$T_{2i} = 317.201 - \frac{V_{2i}^2}{1.01}$$

$$\underbrace{\dot{m}}_{\text{known}} = \frac{P_{2i}}{R\left(317.201 - \frac{V_{2i}^2}{1.01}\right)} A_2 \cdot V_{2i}$$

$V_{2i}$ is non-linearly related to $\dot{m}$.
We also don't know $P_{2i}$. We must measure practically.

★ For example, we have measured $P_{2i} = 1.05 \times 10^5$ Pa using water column, then, $V_{2i} = 2.75$ m/s

$T_{2i} = 301.332$ K

$\rho_{2i} = 1.2141$ kg/m$^3$

Figure 6 Figure Iterative Calculation for Diffuser Inlet Conditions

Once we plot the performance curve of the compressor for the already designed combustion chamber as a partially closed valve, we will directly use the mass flow rate and pressure ratio relation obtained from curve fitting of those plots.

We also measure the pressure P2i using the water column and use that in the calculations done above.

## 4.3 Combustor State Variables calculations:

**All equations below are *per kg of air* unless mass flow appears.**

### Useful relations

- **Speed of sound:** *a = sqrt(γ R T)*
- **Flight speed:** *u0 = M0 * sqrt(γ R T0)*
- **Stagnation enthalpy for perfect gas:** *h_t = cp·Tt*.

### Combustor: determine fuel/air ratio f

Energy balance across combustor (steady, per kg air), neglecting pressure loss:

$$f \ \mathrm{LHV}\ \eta_b \ = \ c_p\,(T_{t4} - T_{t3})$$

So

$$\boxed{f \ = \ \frac{c_p\,(T_{t4} - T_{t3})}{\eta_b\,\mathrm{LHV}}}$$

**Note**: *Tt4* is usually chosen (design limit / turbine-inlet temp).

$$T_{t4} - T_{t3} \ = \ \frac{f\,\eta_{comb}\,LHV}{(1+f)\,c_p}$$

For small $f$ $(f \ll 1)$:

$$T_{t4} - T_{t3} \approx \frac{f\,\eta_{comb}\,LHV}{c_p}$$

So increase in $T_{t4}$ is directly proportional to fuel mass flow $\dot{m}_f = f\dot{m}_a$.

### Total-pressure change across combustor (practical model)

Combustor causes *loss* of total pressure (friction, swirl, mixing). Model with burner total-pressure ratio $\pi_b$:

$$\boxed{p_{t4} \ = \ \pi_b\,p_{t3}, \qquad 0 < \pi_b \le 1}$$

(If no losses $\pi_b = 1$; typical values are $0.95 - 0.98$ depending on design.)

## 4.4 Isentropic expansion in nozzle (to ambient $p_0$)

For an isentropic (reversible, adiabatic) expansion from *pt4* to *p_e = p0*:

$$\frac{T_e}{T_{t4}} = \left(\frac{p_e}{p_{t4}}\right)^{(\gamma-1)/\gamma} \quad \Rightarrow \quad \boxed{T_e = T_{t4}\,(\frac{p_0}{p_{t4}})^{(\gamma-1)/\gamma}}$$

Exit velocity from energy eqn:

$$\boxed{u_e = \sqrt{2\,c_p\,(T_{t4} - T_e)}}$$

From isentropic relation between total and static pressures:

$$\frac{p_e}{p_{t4}} = (1 + \gamma - 1/2\,M_e^2)^{-\gamma/(\gamma-1)}$$

Exit Mach number:

$$M_e = \sqrt{\frac{2}{\gamma - 1}\left(\frac{T_{t4}}{T_e} - 1\right)}$$

Or Exit velocity in Mach form:

$$\boxed{u_e \;=\; M_e\sqrt{\gamma R T_e}}$$

Both are identical (since $c_p = \frac{\gamma R}{\gamma - 1}$).

## 4.5 Thrust & specific thrust (steady, 1-D)

If inlet velocity $V_0$ is nonzero (ram effect), general thrust is

$$\boxed{F \;=\; \dot{m}\,u_e - \dot{m}_a\,V_0 + (p_e - p_a)\,A_e}$$

- If engine is static intake $V_0 \approx 0$, the second term drops.
- If perfectly expanded $p_e = p_a$, pressure term $(p_e - p_a)A_e$ = 0 and

$$\boxed{F \;=\; \dot{m}\,u_e \quad (\text{if } p_e = p_a \text{ and } V_0 \approx 0)}$$

Remember $\dot{m} = \dot{m}_a(1 + f)$.

Assuming fully expanded nozzle (no pressure thrust term):

$$\boxed{\frac{F}{\dot{m}_a} = (1+f)\,u_e - u_0}$$

Total thrust:

$$\boxed{F = \dot{m}_a\,[(1+f)\,u_e - u_0]}$$

Fuel mass flow:

$$\dot{m}_f = f \cdot \dot{m}_a$$

### Specific thrust (per kg air)

Same as above: $F/\dot{m}_a = (1+f)u_e - u_0$.

### Fuel/air ratio from combustor specification

$$\boxed{f = \frac{c_p\,(T_{t4} - T_{t3})}{\eta_b\,LHV}}.$$

### Exit velocity expressed via design variables

Substituting, $T_{t3} = T_{t2}\pi_c^{(\gamma-1)/\gamma}$ and $T_{t2} = T_{t0}$:

$$T_{t4}\text{ (given)} \quad \Rightarrow \quad u_e = \sqrt{2c_p\left(T_{t4} - T_{t4}(\frac{p_0}{p_{t4}})^{(\gamma-1)/\gamma}\right)}$$

with $p_{t4} = p_{t3} = p_{t2}\pi_c$ and $p_{t2} = p_{t0}$ from freestream, so $p_{t4} = p_{t0}\,\pi_c$. Thus $u_e$ can be written fully in terms of $T_{t4}$, $T_{t0}$, $\pi_c$, $p_0$.

More compact when you define:

$$q_t \equiv \frac{T_{t4}}{T_0}, \qquad \Rightarrow \; u_e = \sqrt{2c_pT_{t4}(1 - (p_0/p_{t0}\,\pi_c)^{(\gamma-1)/\gamma}}).$$

## 4.6 Specific impulse (ideal)

$$\boxed{I_{sp} = \frac{(F/\dot{m}_a)}{(\dot{m}_f/\dot{m}_a)\,g_0} = \frac{(1+f)u_e - u_0}{f\,g_0}}$$

For small f, approx. $I_{sp} \approx \frac{u_e - u_0}{f g_0}$.

## 4.7 Mass-flow consistency (choked-throat formula)

**Choked condition** (throat Mach = 1) occurs when pressure ratio is sufficient. Critical static/total relation:

$$\left(\frac{p^*}{p_{t4}}\right) = \left(\frac{2}{\gamma + 1}\right)^{\gamma/(\gamma-1)}.$$

Flow is choked at throat if $p_a \le p^*$ (practical check). Equivalently:

$$\text{Choked if} \quad \frac{p_{t4}}{p_a} \ge \left(\frac{\gamma + 1}{2}\right)^{\gamma/(\gamma-1)}.$$

If choked, total mass flow $\dot{m}$ (mixture) through throat:

$$\boxed{\dot{m} \ = \ p_{t4}\, A_t \sqrt{\frac{\gamma}{R\, T_{t4}} \left(\frac{2}{\gamma + 1}\right)^{\gamma+1/2(\gamma-1)}}}$$

Here $R$ is the specific gas constant of the combustion products (mixture).

If not choked, $\dot{m}$ from exit conditions:

$$\dot{m} \ = \ \rho_e\, u_e\, A_e \ = \ \frac{p_e}{RT_e}\, u_e\, A_e$$

and $A_e$ ties to $A_t$ by the area–Mach relation (next).

### Area–Mach relation (get $A_e/A_t$ for exit area)

$$\boxed{\frac{A_e}{A_t} \ = \ \frac{1}{M_e} (\frac{2}{\gamma + 1} (1 + \gamma - 1/2\, M_e^2))^{\gamma+1/2(\gamma-1)}}$$

This plus $\dot{m}$ formula gives $A_e$ consistent with the throat geometry.

# 5. Problem Statement

## 5.1 Combustion Heat Transfer Modelling

Compute the steady-state wall temperature(s) $T_w$ (possibly split into inner/outer region values) such that, for each radial heat path, the heat entering from hot combustion gas (radiation + convection) equals heat leaving to casing/ambient (radiation + convection). The solved $T_w$ is then used to evaluate material limits, cooling demand, and the maximum permissible gas temperature that sets thrust limits.

### Physical domains & heat paths

- Gas domain (combustion gases at $T_g$): supplies heat by radiation and forced convection to the inner liner surface.
- Wall (liner): thin structural layer; assumed isothermal through thickness for lumped model or discretized if conduction through thickness is important.
- Casing / ambient domain ($T_c, T_a$): receives heat from wall via radiation to casing and convection to ambient air.
- Radial splitting: heat from gas divides through two radial routes (inner and outer region) with area ratios $AR_i$ and $AR_o$ (sum = 1).

## 5.2 Nozzle Design

Design a nozzle that delivers the required thrust by checking if the flow chokes and determining the corresponding Mach numbers, area ratios, and exit conditions. Use choke criteria, static–total relations, and area–Mach equations to identify the correct operating mode and compute exit properties. The goal is to select throat and exit areas that satisfy the pressure ratio and provide the target thrust under ideal, adiabatic, isentropic expansion, while also defining thermal, propulsive, and overall efficiencies.

- The engine suffers from low mass flow rate, resulting in insufficient airflow for producing higher thrust.
- Thrust output is low because the compressor cannot generate adequate pressure rise or flow momentum.
- RPM is limited by the fact that the system is driven only by a motor shaft; There is no turbine to naturally accelerate the shaft to high speeds.
- Low RPM leads to low compression pressure ratio, which directly restricts overall performance.

- The system experiences vibrations and mechanical instability during operation.
- The impeller–diffuser clearance is very high, reducing compressor efficiency.
- When attempting to decrease the clearance, the impeller touches the diffuser:
    - This causes severe vibrations,
    - Mechanical wear or damage,
    - And makes high-RPM operation impossible.
- Due to these combined aerodynamic and mechanical constraints, achieving the required mass flow, pressure rise, and thrust becomes impossible in the current configuration.

# 6. Design and Methodology

## 6.1 COMPONENT DIMENSION DESIGN:

### Mathematical Relations and Formulae :

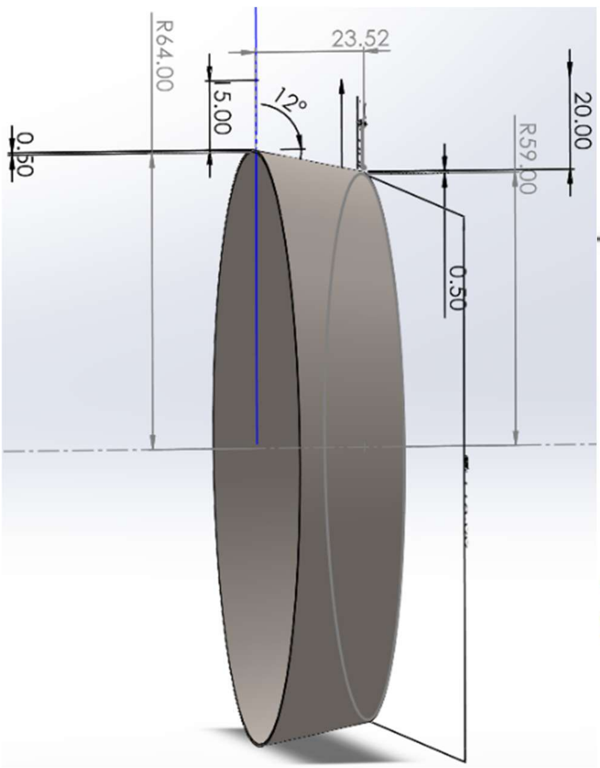
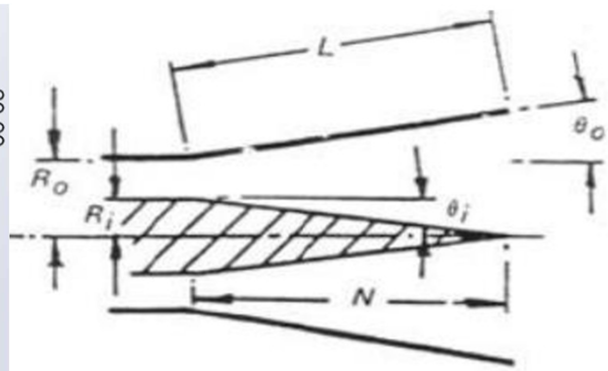



*Figure 7 Fabrication Diffuser using Sheet metal*



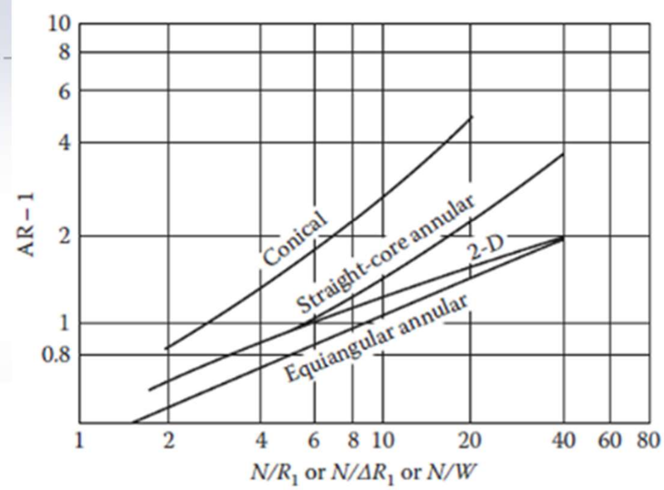



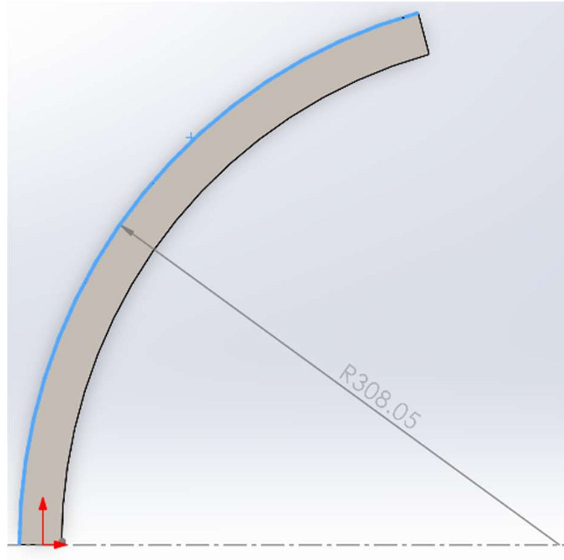

Figure 2 Lines of First Stall

### I. Diffuser Sizing

Equiangular Diffuser Aspect Ratio ($AR$):

$$AR = 1 + 2\frac{L}{\Delta R}\frac{\sin\theta_i + (R_i/R_o)\sin\theta_o}{1 + R_i/R_o} + \frac{(L^2/\Delta R^2)(1 - R_i/R_o)(\sin^2\theta_i - \sin^2\theta_o)}{1 + R_i/R_o}$$

Where: - $L$ = Axial length - $\Delta R = R_o - R_i$ (Radial gap) - $\theta_i, \theta_o$ = Inner and Outer divergence angles - $R_i, R_o$ = Inner and Outer inlet radii




| Component | Parameter | Calculated Value | Unit |
|---|---|---|---|
| **Diffuser** | Axial Length ($L$) | **23.52** | mm |
| | Inner Divergence Angle ($\theta_i$) | **12** | deg |
| | Outer Divergence Angle ($\theta_o$) | **0** | deg |
| | Inner Radius ($R_i$) | **59** | mm |
| | Outer Radius ($R_o$) | **64** | mm |
| | Aspect Ratio ($AR$) | **2.06** | - |

Non-dimensional Length:

$$N = \frac{L}{\Delta R}$$

## II. Reference Values

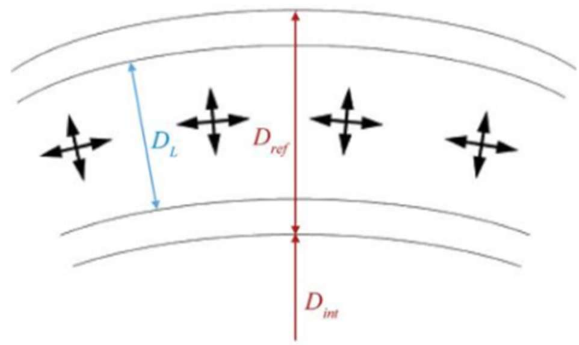



Figure 3 Geometry of Reference Area

Reference Area ($A_{ref}$):

$$A_{ref} = \frac{\pi}{4}\left[\left(2D_{ref} + D_{int}\right)^2 - (D_{int})^2\right]$$

*($D_{ref}$ is typically the casing diameter, $D_{int}$ is the inner diameter)*

Reference Velocity ($U_{ref}$) & Dynamic Pressure ($q_{ref}$):

$$U_{ref} = \frac{\dot{m}_a}{\rho_{t3} A_{ref}}$$

$$q_{ref} = \frac{1}{2}\rho_{t3} U_{ref}^2$$

| Component | Parameter | Calculated Value | Unit |
|---|---|---|---|
| Reference | Reference Area ($A_{ref}$) | $2.26 \times 10^{-3}$ | $m^2$ |
| | Outer Diameter ($D_{out}$) | 79.50 | mm |
| | Inner Diameter ($D_{in}$) | 58.66 | mm |

# III. Pressure Loss Parameters

Total Pressure Loss Ratio:

$$\frac{\Delta p_{t,3-4}}{q_{ref}} = \frac{\Delta p_{t,diff}}{q_{ref}} + \frac{\Delta p_{t,L}}{q_{ref}}$$

Diffuser Pressure Loss:

$$\Delta p_{t,diff} = q_{ref}\lambda\left(1 - \frac{1}{AR^2}\right)$$

*(Where $\lambda$ is the dump diffuser loss coefficient)*

Figure 8 Cross section of liner and casing

# IV. Liner Sizing & Length

Liner Area ($A_L$):

$$A_L = kA_{ref}$$

[cite: 824] *(Where k is a constant, typically 0.66 – 0.70)*

Liner Diameter / Flame Tube Height ($D_L$):

$$D_L = \frac{D_{L,o} - D_{L,i}}{2}$$

Pattern Factor ($PF$):

$$PF = \frac{T_{t,max} - T_{t4}}{T_{t4} - T_{t3}}$$

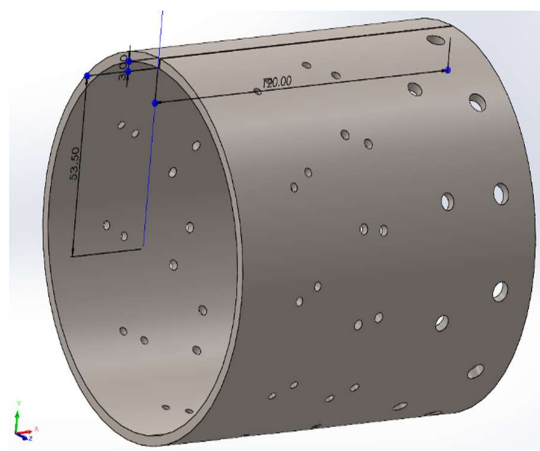


Figure 9 Outer Liner

**Total Liner Length ($L_L$):**

$$L_L = D_L\left(A\frac{\Delta p_{L,L}}{q_{ref}}\ln\frac{1}{1-PF}\right)^{-1}$$

| Component | Parameter | Calculated Value | Unit |
|---|---|---|---|
| Liner Sizing | Liner Area ($A_L$) | $1.583 \times 10^{-3}$ | $m^2$ |
| | Flame Tube Height ($D_L$) | 10.42 | mm |
| | Outer Liner Diameter ($D_{L,o}$) | 79.50 | mm |
| | Inner Liner Diameter ($D_{L,i}$) | 58.66 | mm |

## V. Zone Lengths (Rule of Thumb)

- Primary Zone: $L_{PZ} = \frac{2}{3}D_L$ to $\frac{3}{4}D_L$
- Secondary Zone: $L_{SZ} = \frac{1}{2}D_L$
- Dilution Zone: $L_{DZ} = L_L - L_{PZ} - L_{SZ}$

| Component | Parameter | Calculated Value | Unit |
|---|---|---|---|
| Zone Lengths | Total Liner Length ($L_L$) | 125 | mm |
| | Primary Zone Length ($L_{PZ}$) | 7.82 | mm |
| | Secondary Zone Length ($L_{SZ}$) | 5.21 | mm |
| | Dilution Zone Length ($L_{DZ}$) | 111.97 | mm |

## VI. Airflow Distribution

- Primary Zone Mass Flow: $\dot{m}_{PZ} = 14.77 \times \alpha_{PZ} \times \dot{m}_f$ `
- Secondary Zone Mass Flow: $\dot{m}_{SZ} = 14.77 \times \alpha_{SZ} \times \dot{m}_f$
- Dilution Zone Mass Flow: $\dot{m}_{DZ} = \dot{m}_a - (\dot{m}_{PZ} + \dot{m}_{SZ})$
- Liner Hole Sizing

Mass Flow per Hole:

$$\dot{m}_j = \frac{\pi}{4} n d_j^2 \rho_{t3} U_j$$

Jet Velocity ($U_j$):

$$U_j = \sqrt{\frac{2\Delta p_{t,L}}{\rho_{t3}}}$$

a. Penetration Depth ($Y_{max}$):

$$\frac{Y_{max}}{d_j} = 1.25 \frac{\dot{m}_a}{\dot{m}_a + \dot{m}_j} \sqrt{\frac{\rho_j U_j^2}{\rho_g U_g^2}}$$

b. Actual Hole Diameter ($d_h$):

$$d_h = \frac{d_j}{\sqrt{C_D}}$$

*(Where $C_D$ is discharge coefficient, typically 0.6)*

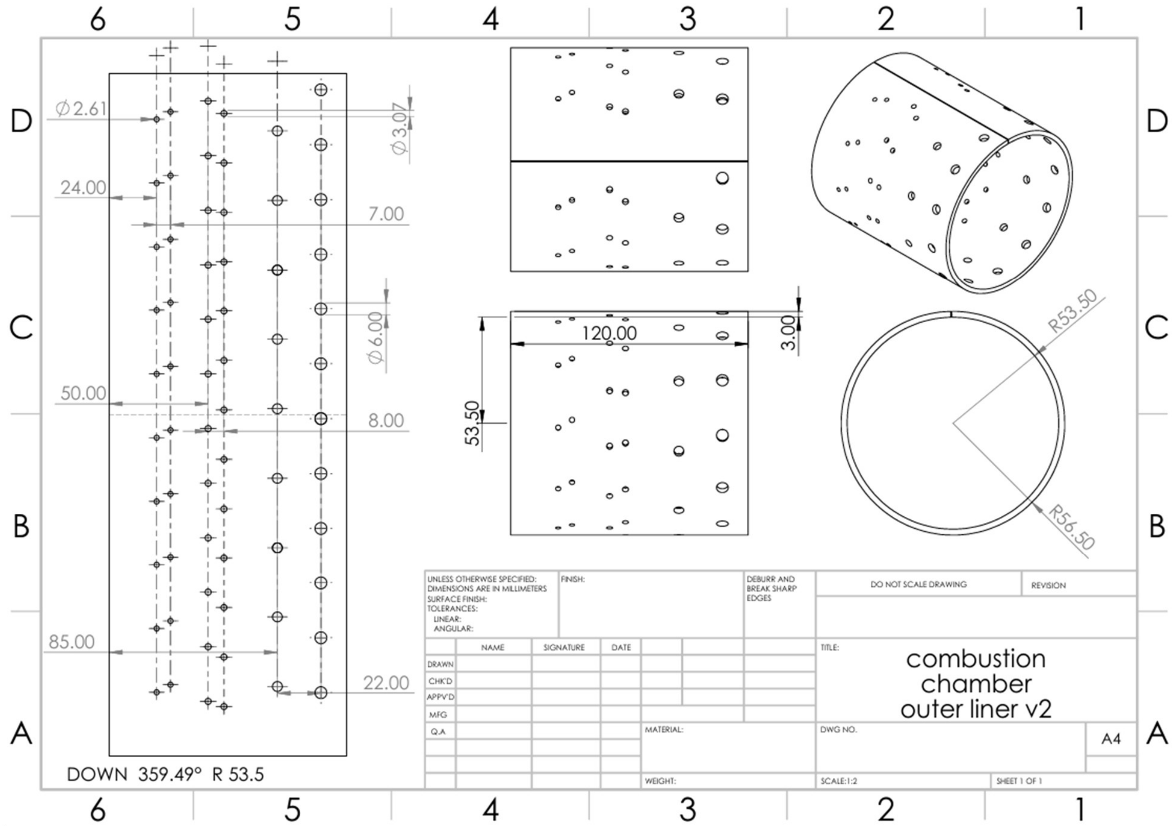



Figure 10 Outer Liner Drawing

| Component | Parameter | Calculated Value | Unit |
|---|---|---|---|
| Hole Distribution | Primary Zone (Axial) | 2.61 | mm (12 holes) |
| | Primary Zone (Annular) | 2.61 | mm (12 holes) |
| | Secondary Zone | 3.07 | mm (10 holes/row) |
| | Dilution Zone | 5.50 | mm (8 holes) |

## VII. Injector Placement Criteria

- **Angular Spread** ($\theta$): 32.72° to 360° (increments of $\sim 32.73°$) for symmetry.
- **Distances from Liner Wall:** Horizontal ($H$): 34–40 mm; Vertical ($V$): 35–43 mm.
- **Goal:** Place injectors where recirculation strength provides ~50% reversal velocity to maximize mixing and flame holding.

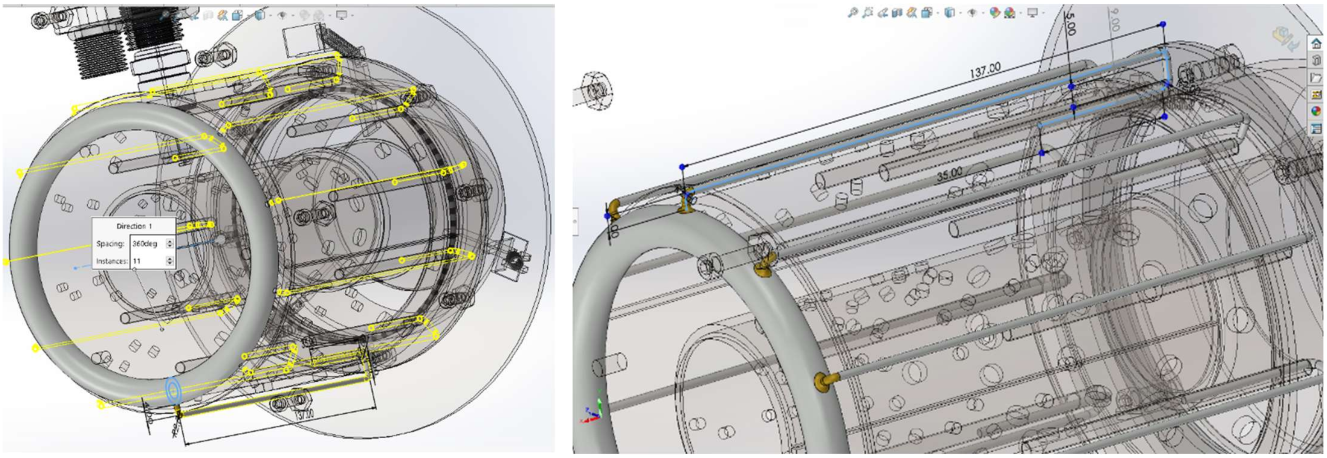
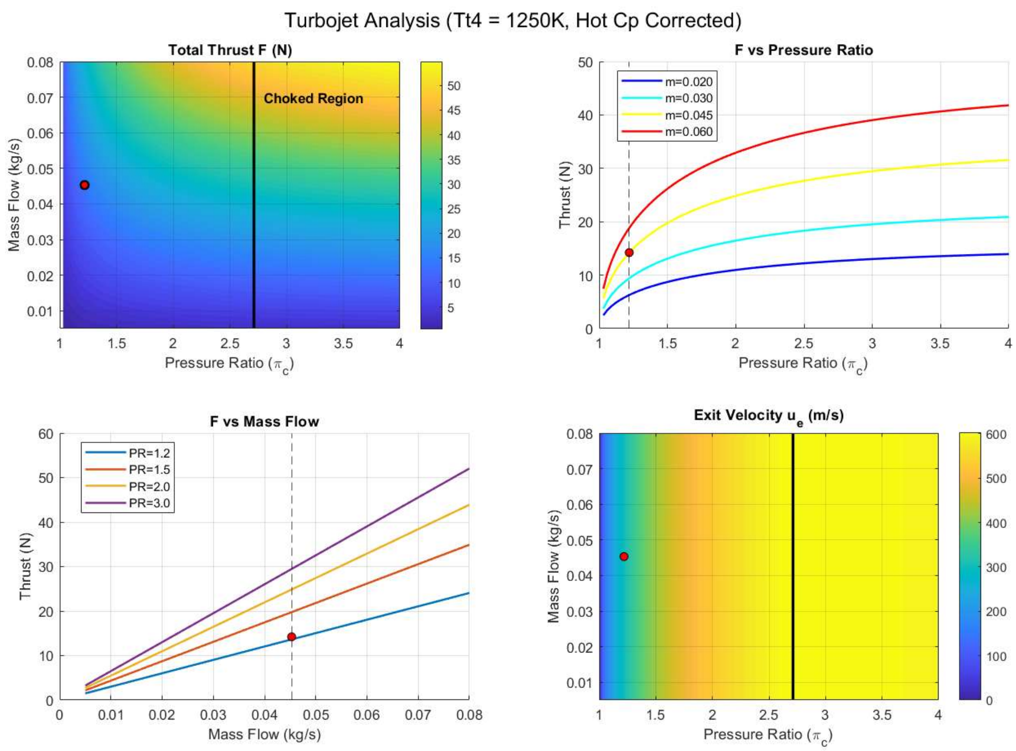



*Figure 11 Position of Injection tubes going over outer liner, evaporating fuel from heat of outer liner before injection*

## 6.2 Graphs:




### 1. Thrust vs. Mass Flow (Linear Trend)

Thrust increases in a perfectly straight line with mass flow because pushing twice as much air generates exactly twice as much force. This means the easiest way to scale up your engine's power is simply to make it bigger (larger diameter) to swallow more air.

## 2. Thrust vs. Pressure Ratio (Diminishing Returns)

Thrust rises sharply at low pressure ratios but eventually flattens out because the energy cost to drive the compressor starts to outweigh the benefits. You get a huge boost early on, but eventually, squeezing the air harder yields very little extra power.

## 3. The Choked Flow Boundary (Vertical Line at $(\pi c = 2.75)$)

The nozzle only hits the speed of sound (chokes) at a Pressure Ratio of $\pi c = 2.75$, which is much higher than a simple tank. This is because the turbine "steals" most of the generated pressure to drive the compressor, leaving only the leftovers to accelerate the exhaust.

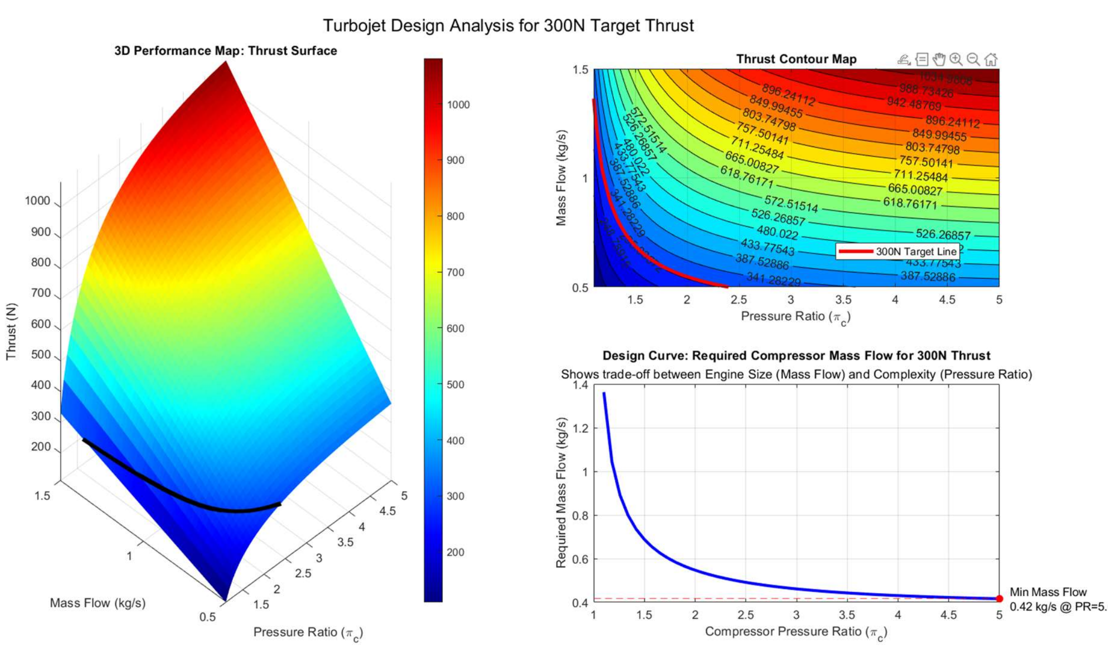
## 4. Independence of Exit Velocity (Vertical Color Bands)

The vertical color bands show that exhaust speed depends only on the Pressure Ratio, not on the Mass Flow. You cannot make the exhaust stream faster by making the engine bigger; you must increase the pressure or temperature.

## 5. Baseline Design Point (The Red Dot)

Your design point sits in the blue "Unchoked" region, meaning the engine operates at subsonic speeds (like a powerful fan) rather than a supersonic jet. It prioritizes mechanical simplicity (low stress) over high thrust efficiency.

1. Thrust Dependence (3D Surface & Contour): Thrust increases linearly with Mass Flow (engine size) but shows diminishing returns with increased Pressure Ratio




(cycle complexity). The black/red target lines trace the specific combinations of these two variables that achieve exactly 300N of thrust.

**2. The Size-Efficiency Trade-off (Bottom Right):** To maintain constant thrust (300N), increasing the Pressure Ratio allows you to significantly decrease the required Mass Flow. For example, a low-pressure fan ($\pi c \sim 1.2$) requires a massive **1.3 kg/s** of air, while a high-pressure compressor ($\pi c \sim 5$) needs only **0.42 kg/s**.

**3. Diminishing Returns on Complexity:** The design curve flattens out after ($\pi c \sim 3.0$), meaning that adding more compressor stages beyond this point yields very little reduction in engine size (mass flow) while adding significant mechanical complexity and weight.

**4. Design Point Selection:** For a micro-jet, the "sweet spot" is likely between **PR 2.0 and 3.0**, where the required mass flow drops to a manageable level (~0.6 kg/s) without requiring the heavy multi-stage compressor needed for PR 5.0.

**5. Physical Significance:** This analysis proves that you can achieve your 300N target either by building a **large, simple engine** (high mass flow, low pressure) or a **small, complex engine** (low mass flow, high pressure), guiding your choice based on manufacturing constraints.

## 6.3 Heat transfer model used for the liner wall temperature calculations

### Under steady state conditions:

Heat entering wall from gas = Heat leaving wall to surroundings

$R_1$ – Radiation from hot gas to inner wall

$$R_1 = \frac{1}{2}\sigma(1+\varepsilon_w)\varepsilon_g T_g^{1.5}\left(T_g^{2.5} - T_w^{2.5}\right)$$

This is the radiative heat transfer rate from hot combustion gases to the wall surface.

$C_1$ – Convection from hot gas to inner wall

$$C_1 = 0.017\left(\frac{k_g}{\mu_g^{0.8}}\right)\left(\frac{\dot{m}_g}{A_g}\right)^{0.8} d_h^{-0.2}\left(T_g - T_w\right)$$

This is convective heat transfer from the hot gas to the wall.

$R_2$ – Radiation from wall to outer casing / surroundings

$$R_2 = \frac{\varepsilon_w \varepsilon_c}{\varepsilon_w \varepsilon_c\left(\frac{1}{F_{wc}} - 1\right) + \varepsilon_c + \varepsilon_w(1-\varepsilon_c)}\frac{A_w}{A_c}\sigma(T_w^4 - T_c^4)$$

This is radiation leaving the wall to the outer casing or surroundings.

$C_2$ – Convection from outer wall to ambient air

$$C_2 = 0.02\left(\frac{k_a}{\mu_a^{0.8}}\right)\left(\frac{\dot{m}_a}{A_{sa}}\right)^{0.8} d_{ha}^{-0.2}(T_w - T_a)$$

This is convective heat loss from the outer wall to ambient air.

$$R_1 + C_1 = R_2 + C_2$$

Heat entering wall from gas = Heat leaving wall to surroundings (under steady state conditions)

$$AR_i(R_1 + C_1) + AR_o(R_1 + C_1) = (R_2 + C_2)_i + (R_2 + C_2)_o$$

Here:
- $AR_i$ = fraction of total heat transfer area on inner path
- $AR_o$ = fraction on outer path

$$AR_i + AR_o = 1$$

Total heat = Inner path heat + Outer path heat

$AR_i = \frac{A_{iL}}{A_{iL}+A_{oL}}$ (inner radial direction)

$AR_o = \frac{A_{oL}}{A_{iL}+A_{oL}}$ (outer radial direction)

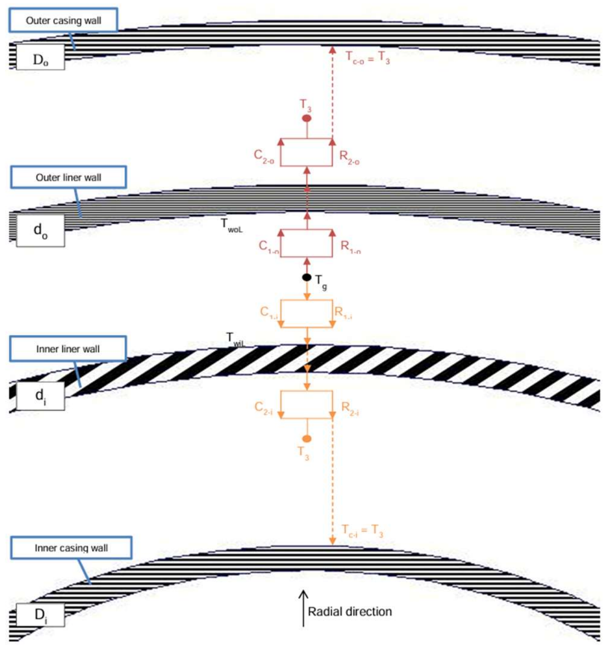



**Figure 6: A diagram of the heat transfer model used for the liner wall temperature calculations**

**Figure 6: A diagram of the heat transfer model used for the liner wall temperature calculations**

$AR_i, AR_o$ are area ratios that the total heat transfer goes through each path.

- $A_{iL}$ = heat transfer area associated with inner region of liner
- $A_{oL}$ = heat transfer area associated with outer region of liner

Inner path balance:

$$AR_i(R_1 + C_1) = (R_2 + C_2)_i$$

Outer path balance:

$$AR_o(R_1 + C_1) = (R_2 + C_2)_o$$

The heat that enters each region must leave that same region.

where,

$$R_1 + C_1 = \left(AT_w^{2.5} - AT_g^{2.5}\right) + \left(BT_w - BT_g\right)$$

Group terms in $T_w$ and the constants:

$$R_1 + C_1 = AT_w^{2.5} + BT_w - AT_g^{2.5} - BT_g$$

Now define

$$C = -AT_g^{2.5} - BT_g$$

So the gas-side term becomes

$R_1 + C_1 = AT_w^{2.5} + BT_w + C$

where,

$A = -\frac{1}{2}\sigma(1+\varepsilon_w)\varepsilon_g T_g^{1.5}, B = -0.017\left(\frac{k_g}{\mu_g^{0.8}}\right)\left(\frac{\dot{m}_g}{A_g}\right)^{0.8} d_{hg}^{-0.2}$

$$R_2 + C_2 = (DT_w^4 - DT_a^4) + (ET_w - ET_a)$$

$$R_2 + C_2 = DT_w^4 + ET_w - DT_a^4 - ET_a$$

$$F = -DT_a^4 - ET_a$$

So the outer-side term becomes

$$R_2 + C_2 = DT_w^4 + ET_w + F$$

Where,

$D = \frac{\varepsilon_w \varepsilon_c}{\varepsilon_w \varepsilon_c\left(\frac{1}{F_{wc}} - 1\right) + \varepsilon_c + \varepsilon_w(1-\varepsilon_c)} \frac{A_w}{A_c}\sigma,\ E = 0.02\left(\frac{k_a}{\mu_a^{0.8}}\right)\left(\frac{\dot{m}_a}{A_{ga}}\right)^{0.8} d_{ha}^{-0.2}$

Take the heat balance for a given radial direction:

$$AR(R_1 + C_1) = R_2 + C_2$$

Substitute the compact forms from sections 2 and 3:

$$AR(AT_w^{2.5} + BT_w + C) = DT_w^4 + ET_w + F$$

Bring everything to one side:

$$AR(AT_w^{2.5} + BT_w + C) - (DT_w^4 + ET_w + F) = 0$$

| Input parameter | Symbol | Value | Units / Notes |
|---|---|---|---|
| Stefan-Boltzmann constant | sigma | 5.67E-08 | W/m^2-K^4 |
| Wall emissivity | eps_w | 0.25 | - |
| Gas emissivity | eps_g | 0.36 | - |
| Gas temperature | Tg | 1500 | K |
| Gas thermal conductivity | kg | 0.16 | W/m-K |
| Gas viscosity | mu_g | 0.0006 | Pa-s |
| Gas mass flow rate | mdot_g | 0.0015 | kg/s |
| Gas flow area | Ag | 0.024492 | m^2 (244.92 cm^2) |
| Gas hydraulic diameter | dhg | 0.065 | m (6.5 cm) |
| Casing emissivity | eps_c | 0.25 | set = eps_w |
| Ambient/cooling air temp | Ta | 300 | K |
| Air thermal conductivity | ka | 0.025 | W/m-K |
| Air viscosity | mu_a | 0.000018 | Pa-s |
| Air mass flow rate | mdot_a | 0.0453 | kg/s |
| Cooling air flow area | Aga | 0.016016 | m^2 (160.16 cm^2) |
| Air hydraulic diameter | dha | 0.044 | m (4.4 cm) |
| View factor liner->casing | Fwc | 1 | - |
| Area ratio Aw/Ac | Aw/Ac | 1.53 | Aw/Ac = 244.92/160.16 |
| | | | |
| | | | |
| Derived coefficients (A–F) | | | |
| A | -0.000741144 | | -0.000741144 |
| B | -0.190159856 | | -0.190159856 |
| C | 64870.08353 | | 64870.08353 |
| D | 1.2393E-08 | | 6.6007E-09 |
| E | 13.40592479 | | 0.877589107 |
| F | -4122.160736 | | -4122.160736 |
| Tw_guess (K) | 1221.14475 | | |
| | 0.000171599 | | |
| | | | |
| Gas side: A*Tw^2.5 + B*Tw + C | 64870.0835 | | |
| Loss side: D*Tw^4 + E*Tw + F | -4122.158435 | | |
| Residual f(Tw) = Gas - Loss | 68992.24194 | | |

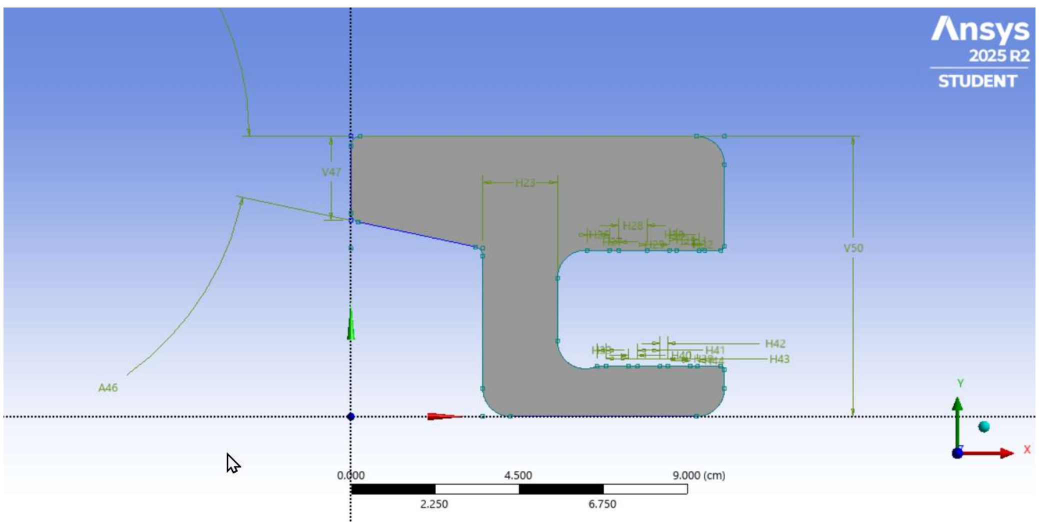
Figure 12 Input parameters, material properties, gas-side/air-side flow characteristics, and derived coefficients used for wall temperature prediction.

## 6.4 CFD Analysis of Diffuser

Airflow From Compressor Exit into Combustion Chamber

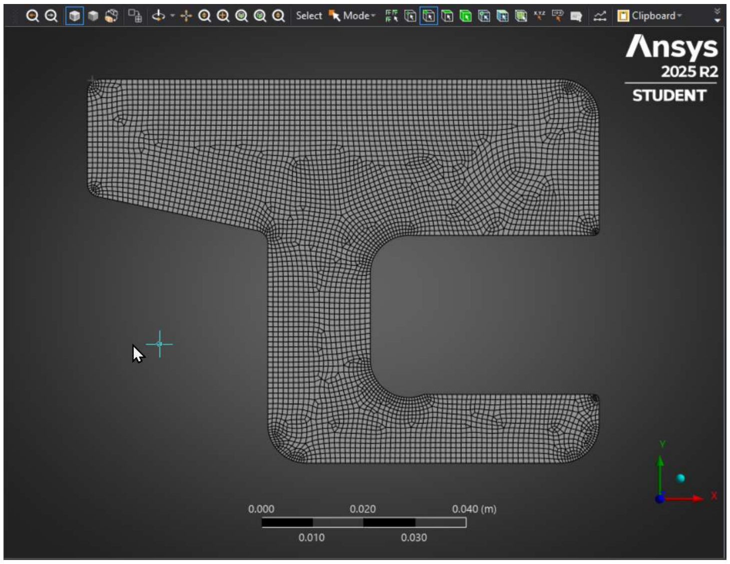
Software: ANSYS Fluent 2025 R2 Student
Turbulence Model: Standard k–ε (two-equation RANS), Double precision.
Flow Regime: Turbulent, steady-state
Working Fluid: Air at ambient inlet conditions
Domain: 2D cross-section representing compressor exit duct leading toward combustion chamber (When combustion is not happening)

### Geometry and Mesh:




The computational domain consists of a duct through which compressed air flows toward the combustion section. The geometry includes curved corners and an internal step that causes flow curvature and local deceleration.

- As it is an axis symmetric problem, we don't have to solve for both the halve




## Mesh Description:

- A structured/grid-like mesh was generated using quadrilateral elements.
- Mesh density is higher (2 times the mess size compared to remaining areas) near the curved internal corners, where adverse pressure gradients and potential flow separation may occur.

1. ## Boundary Conditions:

- Inlet: Uniform velocity profile at ambient conditions (5.5 m/s)
- Outlet: Pressure outlet, static pressure ~0 gauge.
- Walls: No-slip boundary condition (Default)

2. ## Solver:

K-Epsilon (Turbulence model), SIMPLE (Semi-Implicit Method for Pressure-Linked Equations. It's a pressure-velocity coupling scheme (numerical method) for solving the Navier-Stokes equations iteratively).

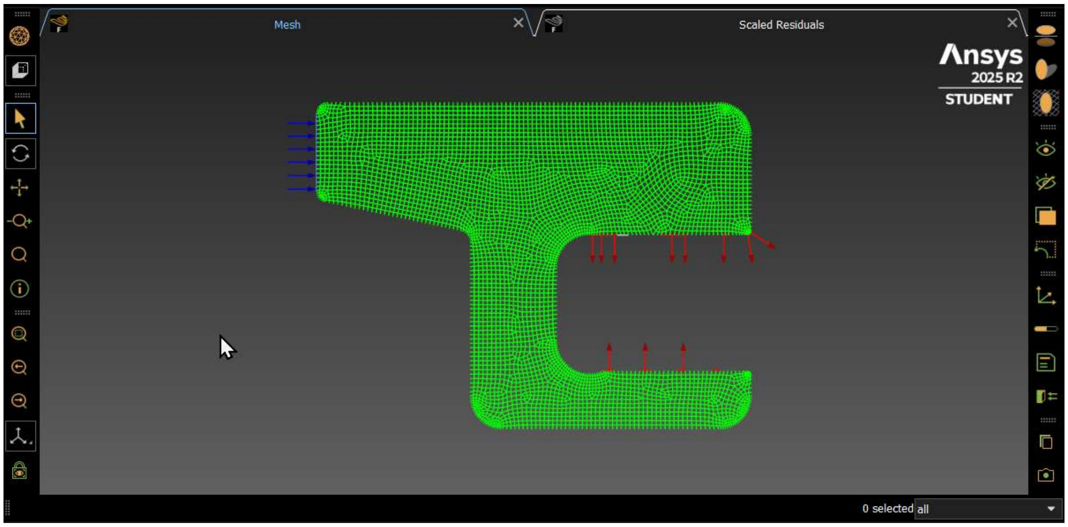
The standard k–ε model is a two-equation Reynolds-Averaged Navier–Stokes (RANS) turbulence model.

The standard k–ε model performs well for:

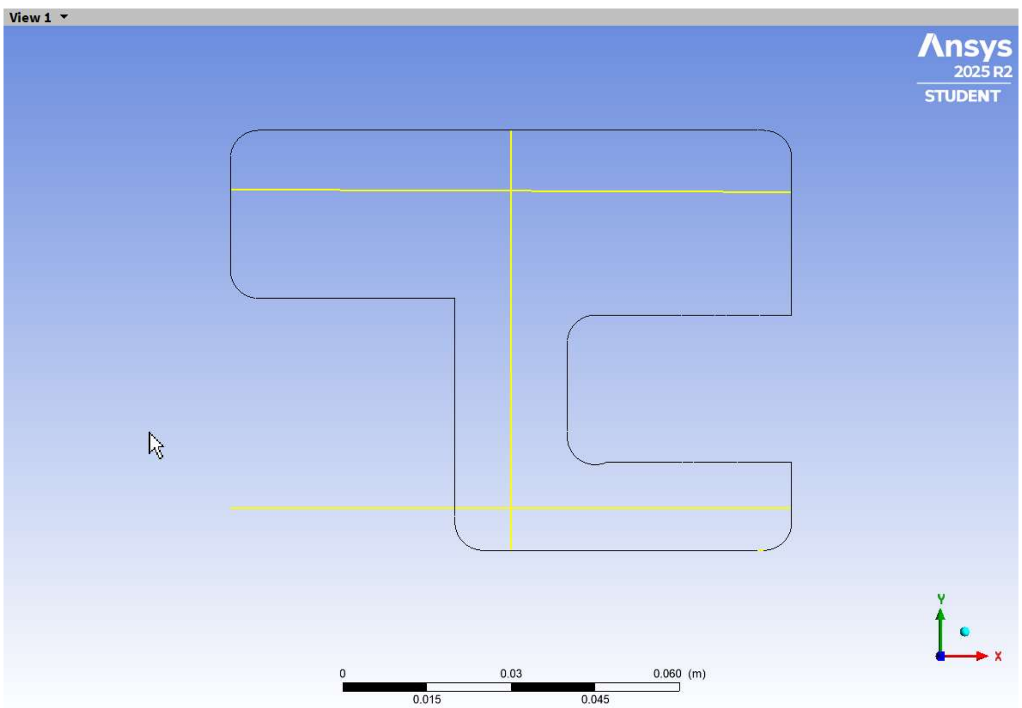
- Duct flows,
- Weak curvature,
- Moderate turbulence levels.

However, k–ε is less accurate for strong adverse pressure gradients or near-wall separation.

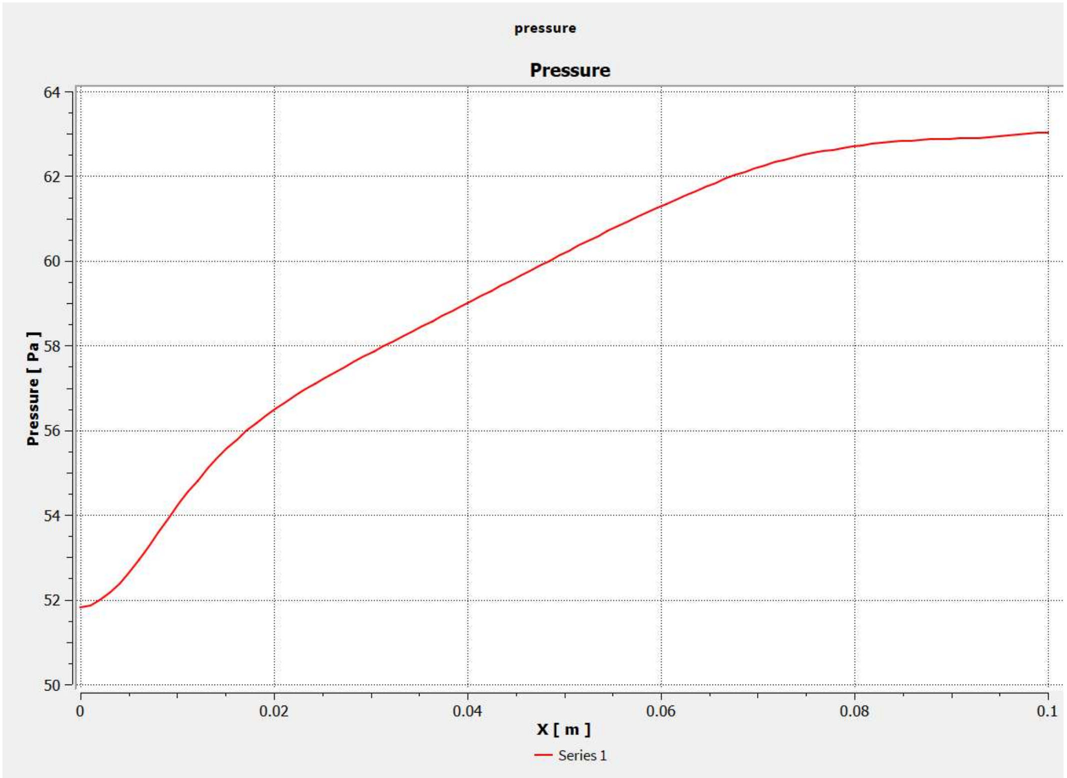
## 3. Flow path of air:

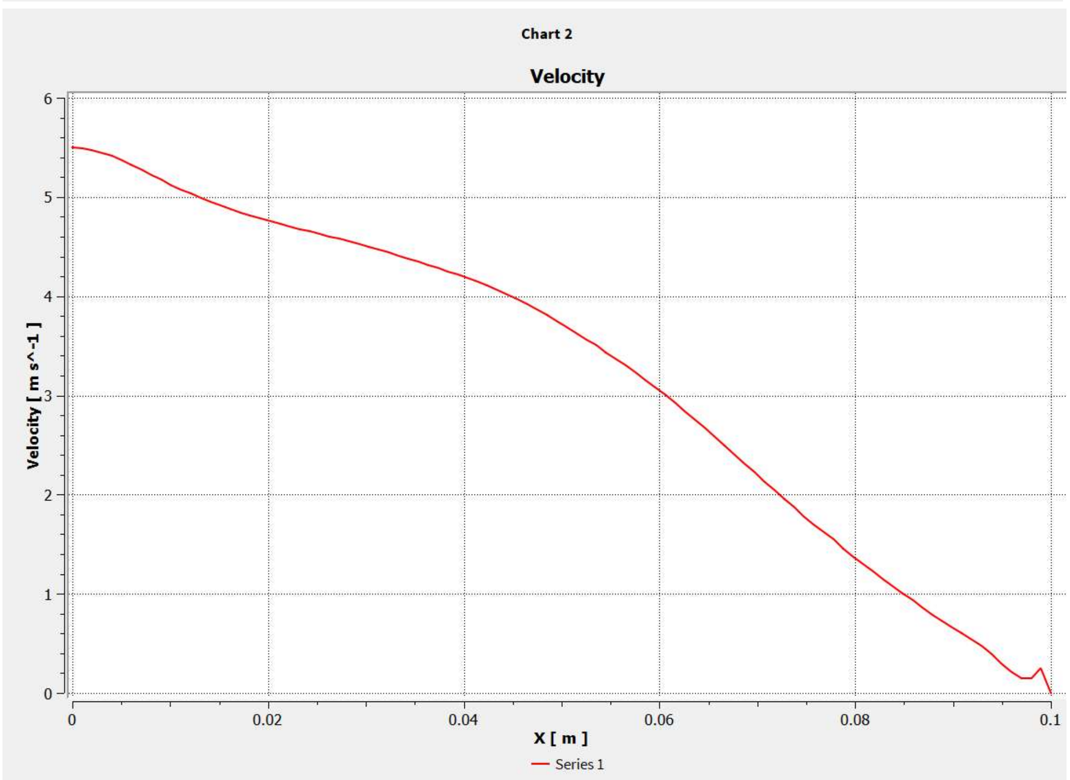



## 4. Results:

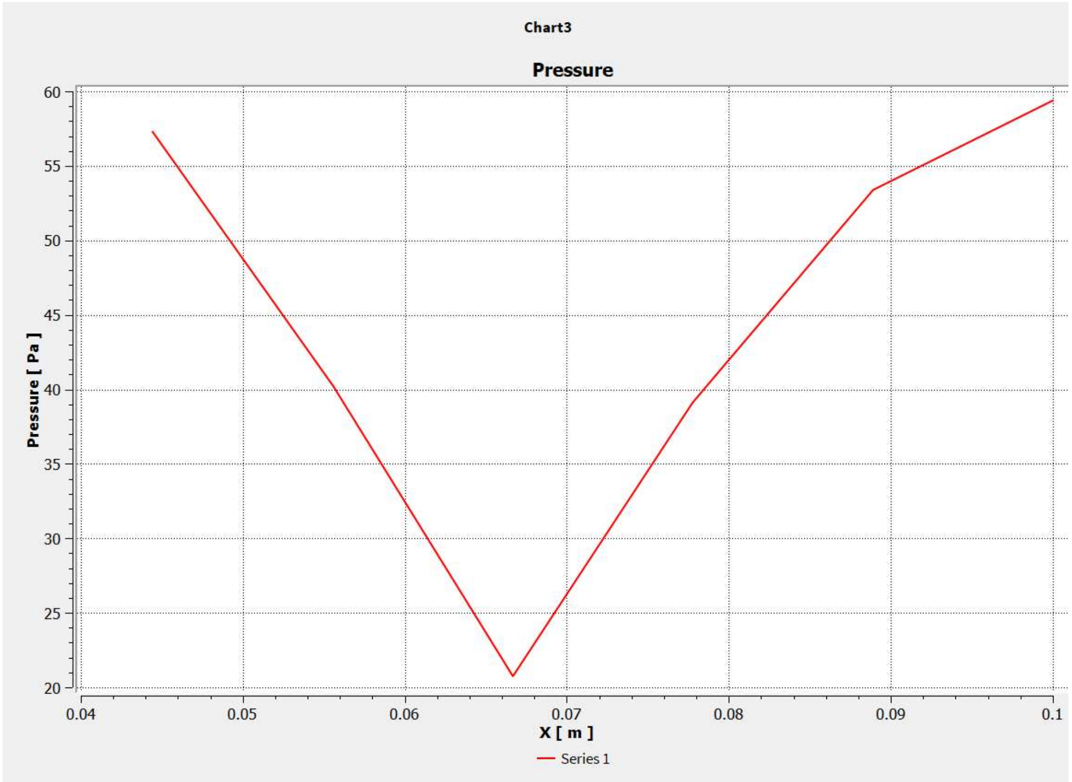


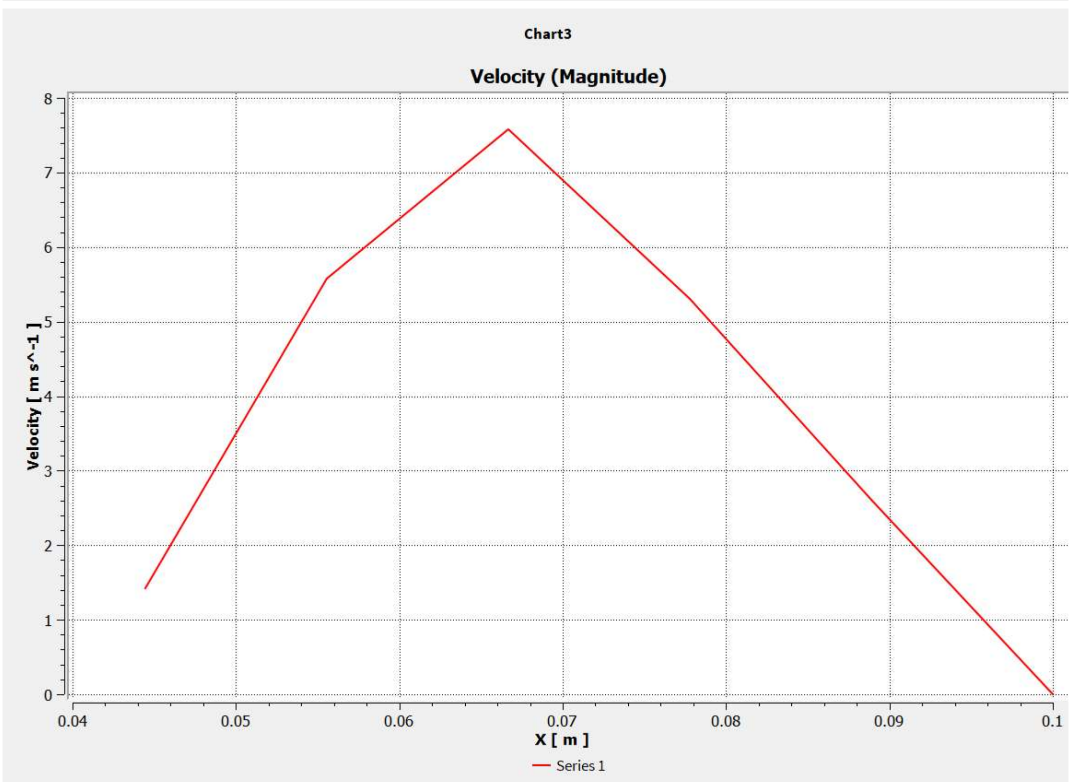
The properties along the top yellow line are as shown:



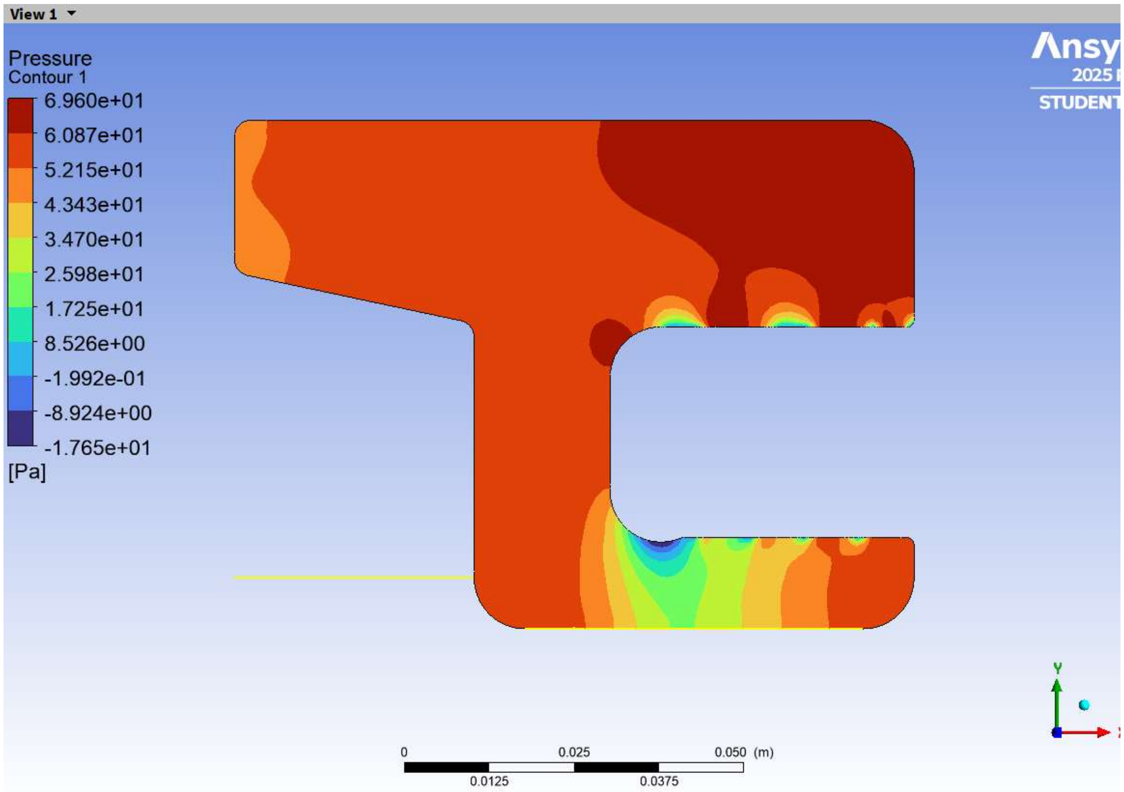



## The properties along the bottom horizontal line:

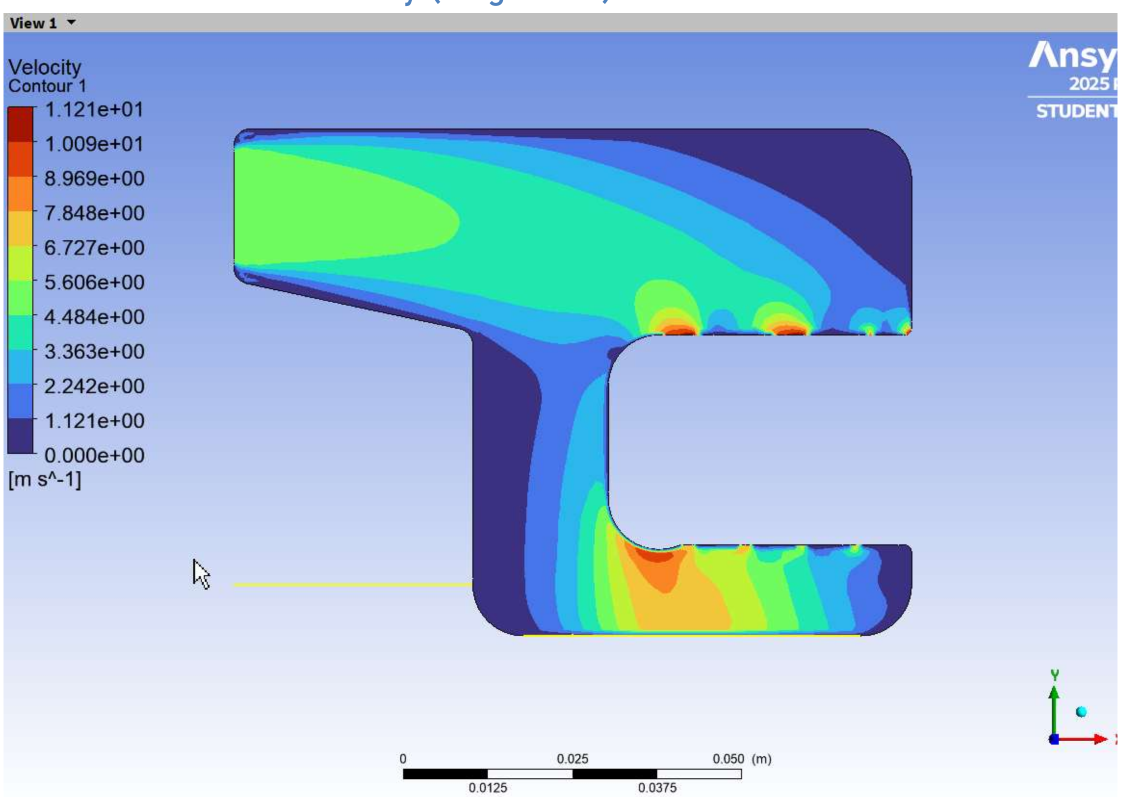


## Overall contour of Pressure (Static):




## Overall contour of Velocity (Magnitude):



### Conclusion:

- As expected, the pressure has built-up across the diverging duct and the velocity has decreased.
- There pressure build up around the outer liner is significantly large compared to the pressure built-up inside the inner liner.
- Though this is just a replica of the actual flow (In the actual flow, combustion will be happening and the pressure won't drop suddenly to atmospheric pressure which is our assumptions here), the above two conclusions are still true even when combustion happens.

# 6.5 NOZZLE DESIGN

## I) Choke criterion (fundamental)

$$\left(\frac{p_e}{p_t}\right)_{\text{crit}} = (\frac{2}{\gamma+1})^{\gamma/(\gamma-1)} \equiv p_{\text{crit}}$$

For air ($\gamma = 1.4$):

$$p_{\text{crit}} \approx 0.528285 \quad \Rightarrow \quad \frac{p_t}{p_0} \gtrsim \frac{1}{p_{\text{crit}}} \approx 1.8925$$

Decision: if $\frac{p_0}{p_t} \le p_{\text{crit}}$ (equivalently $p_t/p_0 \ge 1.8925$) the nozzle *can* choke (sonic at throat) for appropriate geometry.

## II) Static→total conversions (use when inputs are static)

Given $T,\ u$ at a station:

$$T_t = T + \frac{u^2}{2c_p}\, a = \sqrt{\gamma RT}, \quad M = \frac{u}{a}\, p_t = p(1 + \frac{\gamma-1}{2}M^2)^{\gamma/(\gamma-1)}$$

## 1) Unchoked nozzle (no sonic throat)

Condition: $\frac{p_0}{p_t} > p_{\text{crit}}$.

If fully expanded to ambient ($p_e = p_0$):

$$T_e = T_t(\frac{p_0}{p_t})^{(\gamma-1)/\gamma} \quad u_e = \sqrt{2c_p\,(T_t - T_e)}$$

Thrust (general):

$$F = \dot{m}(1+f)u_e - \dot{m}u_0 + (p_e - p_0)A_e$$

If $p_e = p_0$ simplify: $F = \dot{m}(1+f)u_e - \dot{m}u_0$.

## 2) Choked nozzle (sonic throat) — core relations

### Choked mass flow per unit throat area (M=1)

$$\dot{m}'_{\max} = \frac{\dot{m}}{A_t} = p_t\sqrt{\frac{\gamma}{RT_t}}\;(\frac{2}{\gamma+1})^{\frac{\gamma+1}{2(\gamma-1)}}$$

So if $A_t$ known:

$$\dot{m} = A_t\,p_t\sqrt{\frac{\gamma}{RT_t}}\;(\frac{2}{\gamma+1})^{\frac{\gamma+1}{2(\gamma-1)}}$$

### Area–Mach relation (exit Mach from area ratio)

$$\frac{A}{A_*} = \frac{1}{M}(\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right))^{\frac{\gamma+1}{2(\gamma-1)}}$$

Set $A/A_* = A_e/A_t$. Solve numerically for $M_e$ (supersonic root if $A_e/A_t > 1$).

### From Mach to exit properties

$$\frac{p_e}{p_t} = (1+\frac{\gamma-1}{2}M_e^2)^{-\gamma/(\gamma-1)}, \quad \frac{T_e}{T_t} = (1+\frac{\gamma-1}{2}M_e^2)^{-1} \quad u_e = M_e\sqrt{\gamma RT_e} = \sqrt{2c_p\,(T_t - T_e)}$$

### Thrust (choked)

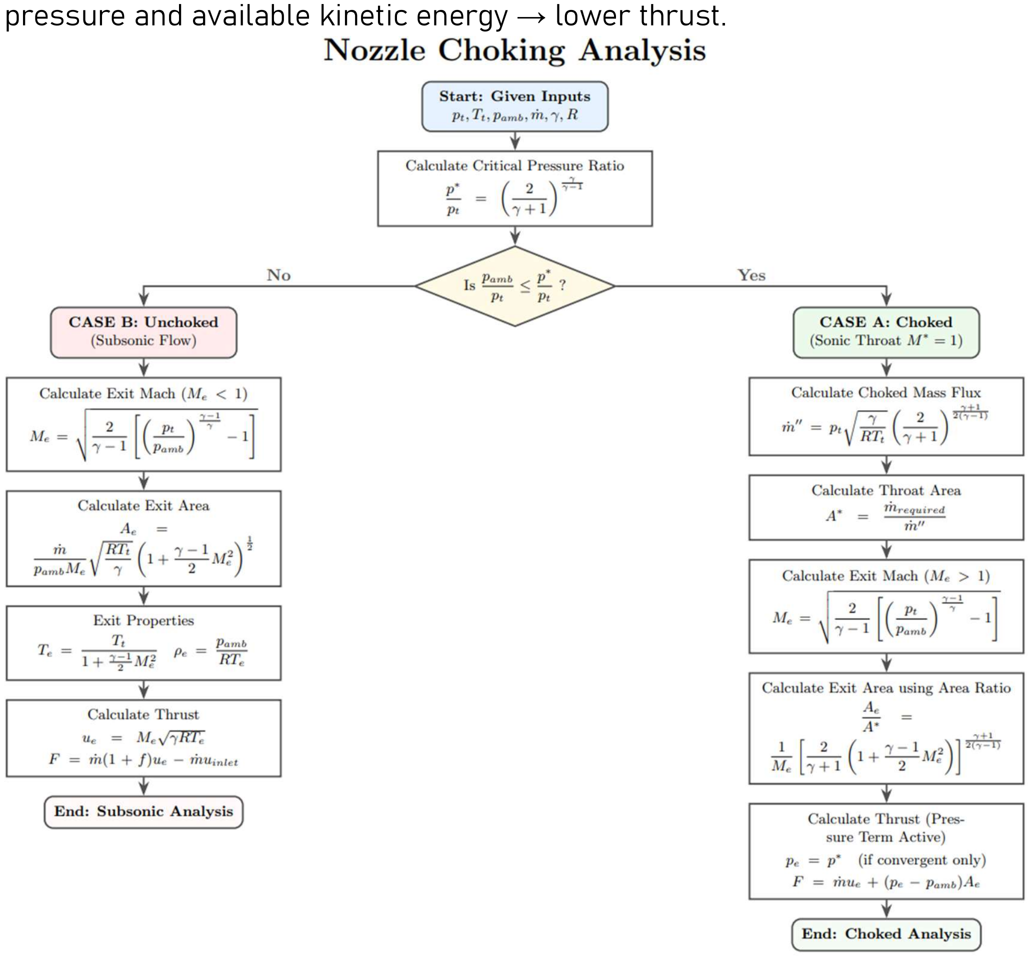
Use same general thrust:

$$F = \dot{m}(1+f)u_e - \dot{m}u_0 + (p_e - p_0)A_e$$

but $\dot{m}$ is from 4.1 and $p_e$ from 4.3. **Note:** $p_e$ need not equal $p_0$.

## Special cases / design targets

- **Choked & ideally-expanded:** choose $A_e/A_t$ such that computed $p_e = p_0$. This yields maximum useful kinetic conversion without external shocks — *usually preferred*.
- Under expanded: $p_e > p_0$ → external expansion fans; pressure thrust term positive $(p_e - p_0)A_e > 0$.
- Overexpanded: $p_e < p_0$ → external shocks; shocks irreversibly reduce total pressure and available kinetic energy → lower thrust.




- *Figure 13 Nozzle Design Approach*

# 6.6 EFFICIENCY:

## Thermal efficiency (engine) — include compressor work penalty

Fuel energy input per kg air: $f\ LHV$. Useful *propulsive* power per kg air:

$$P_{prop} = (1+f)\,\frac{u_e^2 - u_0^2}{2}.$$

If compressor shaft work $W_c$ is provided by the fuel (i.e. internal fuel power pays compressor), net useful power becomes $P_{net} = P_{prop} - W_c$. Thus ideal **thermal efficiency** (net propulsive power divided by fuel chemical power) is:

$$\boxed{\eta_{\text{thermal}} = \frac{(1+f)\ 1/2\,(u_e^2 - u_0^2) - W_c}{f\ LHV}}.$$

With small f and neglect of fuel mass in kinetic term, numerator $\approx 0.5(u_e^2 - u_0^2) - W_c$.

If the compressor work is provided by an external shaft (not from the engine fuel), drop $W_c$ term.

## Propulsive efficiency :

Standard expression (for small f):

$$\boxed{\eta_{\text{prop}} = \frac{\text{propulsive power}}{\text{kinetic power increase}} = \frac{(1+f)u_0(u_e - u_0)}{(1+f)\,1/2\,(u_e^2 - u_0^2)} = \frac{2u_0}{u_e + u_0}}.$$

(This formula is independent of f if the exhaust mass ≈ intake mass.)

## Overall efficiency:

Overall (engine) efficiency = propulsive power / fuel power:

$$\boxed{\eta_{\text{overall}} = \eta_{\text{thermal}} \times \eta_{\text{prop}}}.$$

Or from definitions: $\eta_{\text{overall}} = \frac{(1+f)1/2(u_e^2 - u_0^2) - W_c}{f\ LHV} \times \frac{2u_0}{u_e + u_0}$, the expanded form

- Removing the turbine means **compressor work must be supplied externally** from separate power. If the compressor shaft power comes from the engine fuel, include $W_c$ in the fuel energy budget (as above). For external power the derivation for efficiency is done in later section.
- No diffuser ⇒ stagnation recovery at compressor inlet is just freestream stagnation (no additional pd or td gains).
- If combustor pressure losses exist, set $p_{t4} = p_{t3} \cdot \beta_b$ (β_b<1), and for imperfect expansion negligible term pressure thrust term $(p_e - p_0)A_e$. will be included in further iteration of calculations.

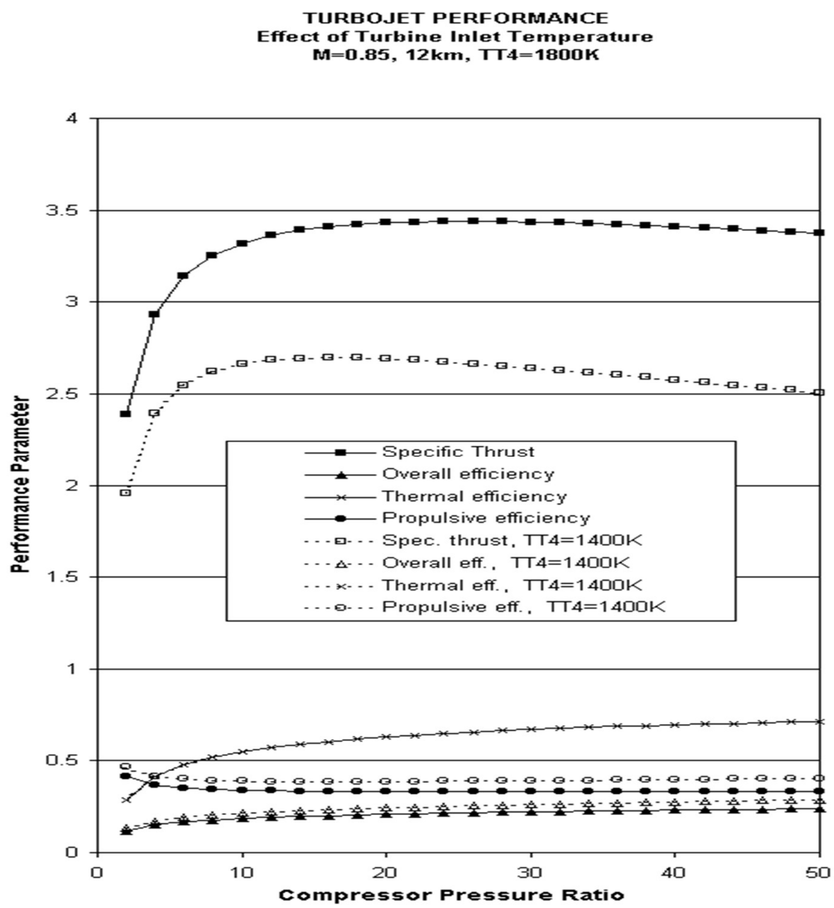



Figure 14 Performance of an ideal turbojet engine as a function of compressor pressure ratio and turbine inlet temperature

# 1) CALCULATION OF PARAMETERS :

## Given / constants / Inputs :

- $p_0 = 100\,000$ Pa
- $V_{2i} = 2.96$ m/s
- $T_{2i} = 308.526$ K
- $\rho_{2i} = 1.13$ kg/m$^3$
- $\dot{m}_a = 0.0453$ kg/s
- $\gamma = 1.4$, $R = 287$ J/(kg · K), $c_p = 1005$ J/(kg · K)
- LHV $= 4.3 \times 10^7$ J/kg, $\eta_b = 0.65$
- $T_{t4} = 1250$ K (stagnation at nozzle inlet)
- $T_{t3} = 331.0713372$ K, $p_{t4} = p_{t3} = 122\,000$ Pa
- $u_0 = 0$ m/s (static test), $g_0 = 9.80665$ m/s$^2$

All calculations assume ideal-gas, isentropic nozzle expansion to ambient $p_e = p_0$

## Critical pressure ratio (choke test)

$$p_{\text{crit}} = (\frac{2}{\gamma + 1})^{\gamma/(\gamma-1)}$$

Numeric:

$$p_{\text{crit}} = (\frac{2}{2.4})^{3.5} \approx 0.528281788.$$

Compare:

$$\frac{p_0}{p_{t4}} = \frac{100000}{122000} \approx 0.819672131 \qquad (0.81967 > 0.52828) \Rightarrow \text{ UNCHOKED (cannot choke).}$$

## Fuel / air ratio $f$ (combustor energy balance)

Formula:

$$f = \frac{c_p\,(T_{t4} - T_{t3})}{\eta_b\ \text{LHV}}.$$

Numeric steps:

$$T_{t4} - T_{t3} = 1250.0000 - 331.0713372 = 918.9286628\ \text{K}.$$

$$f = \frac{1005 \times 918.9286628}{0.65 \times 4.3 \times 10^7} \approx 0.033041979\ \text{(dimensionless)}.$$

Fuel mass flow:

$$\dot{m}_f = f\,\dot{m}_a = 0.033041979 \times 0.0453 \approx 0.001496802\ \text{kg/s}.$$

## Nozzle exit static temperature $T_e$ (isentropic to ambient)

Formula:

$$T_e = T_{t4}(\frac{p_0}{p_{t4}})^{(\gamma-1)/\gamma}.$$

Numeric:

$$\frac{p_0}{p_{t4}} = \frac{100000}{122000} = 0.819672131, \qquad (\gamma - 1)/\gamma = 0.2857142857,$$

$$T_e = 1250 \cdot (0.819672131)^{0.2857142857} \approx \mathbf{1180.961598}\ \text{K}.$$

## Nozzle exit velocity $u_e$ (energy equation),

Formula:

$$u_e = \sqrt{2\, c_p\, (T_{t4} - T_e)}.$$

Numeric:

$$T_{t4} - T_e = 1250 - 1180.961598 = 69.038402 \text{ K},$$

$$2c_p \Delta T = 2 \times 1005 \times 69.038402 \approx 138{,}868.3 \text{ m}^2/\text{s}^2,$$

$$u_e = \sqrt{138{,}868.3} \approx \mathbf{372.51468} \text{ m/s}.$$

## Specific thrust and total thrust:

Specific thrust (per kg air):

$$\frac{F}{\dot{m}_a} = (1 + f)\, u_e - u_0.$$

Numeric:

$$(1 + f)u_e = (1 + 0.033041979) \times 372.51468 \approx 384.8 \text{ N s/kg}.$$

Total thrust:

$$F = \dot{m}_a[(1 + f)u_e - u_0] = 0.0453 \times 384.82330 \approx \mathbf{17.432} \text{ N}.$$

## Specific impulse (ideal)

$$I_{sp} = \frac{(1 + f)u_e - u_0}{f\, g_0}.$$

Numeric:

$$I_{sp} \approx \frac{384.82330}{0.033041979 \times 9.80665} \approx \mathbf{26207.65} \text{ s}.$$

(Note: very large numeric value due to small fuel fraction and small engine scale — consistent mathematically.)

## Required exit Mach $M_e$ for perfect expansion $p_e = p_0$,

From isentropic pressure–Mach relation:

$$\frac{p_0}{p_{t4}} = (1 + \gamma - 1/2\, M_e^2)^{-\gamma/(\gamma-1)} \Rightarrow M_e = \sqrt{\frac{2}{\gamma-1}\left(\left(\frac{p_{t4}}{p_0}\right)^{(\gamma-1)/\gamma} - 1\right)}.$$

Numeric:

$$\frac{p_{t4}}{p_0} = 1.22, \quad (p_{t4}/p_0)^{(\gamma-1)/\gamma} = 1.22^{0.285714} \approx 1.0727,$$

$$M_e = \sqrt{\frac{2}{0.4}\,(1.0727 - 1)} = \sqrt{5 \times 0.0727} \approx$$
$\mathbf{0.54064536}$ (subsonic — consistent with UNCHOKED).

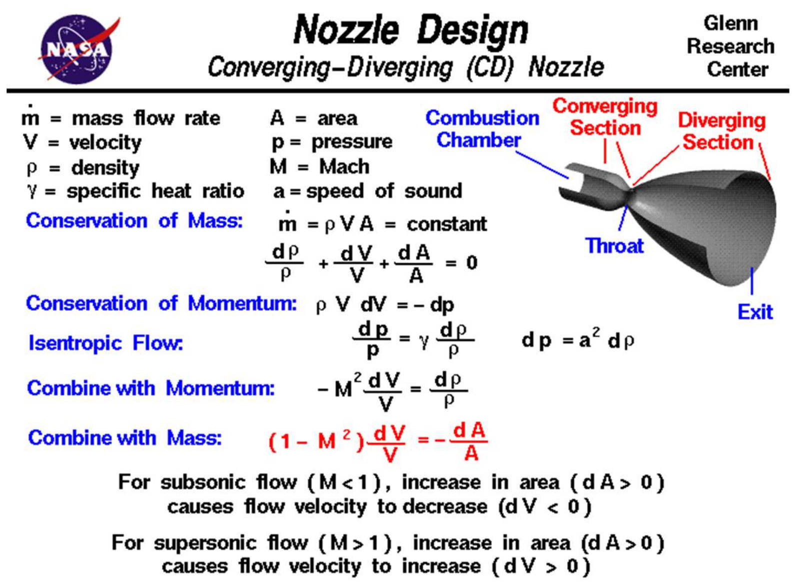
## Area–Mach relation → exit area ratio $A_e/A^*$:

Formula:

$$\frac{A_e}{A^*} = \frac{1}{M_e}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2}M_e^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}.$$

Numeric (with $M_e \approx 0.54064536$):

$$\frac{A_e}{A^*} \approx \mathbf{1.269306472}.$$




## Sonic (critical) area $A^*$ from mass flow:

Inverted choked mass-flux relation:

$$A^* = \frac{\dot{m}_a}{p_{t4}}\sqrt{\frac{RT_{t4}}{\gamma}}\left(\frac{2}{\gamma+1}\right)^{-\frac{\gamma+1}{2(\gamma-1)}}.$$

Numeric:

$$A^* \approx \mathbf{5.52813 \times 10^{-5}}\ \mathrm{m}^2.$$

## Required nozzle exit area $A_e$

$$A_e = A^* \cdot \frac{A_e}{A^*}$$

Numeric:

$$A_e = 5.52813 \times 10^{-5} \times 1.269306472 \approx \mathbf{7.01689 \times 10^{-5}}\ \mathrm{m}^2.$$

Alternate check (continuity):

$$\rho_e = \frac{p_0}{RT_e}, \quad A_e = \frac{\dot{m}_a}{\rho_e u_e}.$$

Using $T_e$ and $u_e$ above gives the same $A_e$ (numerically $\approx 7.01689 \times 10^{-5}\ \mathrm{m}^2$) — consistent.

## Compact results table (rounded)

- $p_{\text{crit}} = 0.528281788$ — UNCHOKED (since $p_0/p_{t4} \approx 0.81967$).
- $f = 0.033041979$
- $\dot{m}_f = 0.001496802\ \mathrm{kg/s}$
- $T_e = 1180.961598\ \mathrm{K}$
- $u_e = 372.51468\ \mathrm{m/s}$
- Specific thrust $= 384.82330\ \mathrm{N/(kg/s)}$
- $F = 17.43250\ \mathrm{N}$
- $I_{sp} = 26207.65\ \mathrm{s}$
- $M_e = 0.54064536$
- $A_e/A^* = 1.269306472$
- $A^* = 5.52813 \times 10^{-5}\ \mathrm{m}^2$
  $A_e = 7.01689 \times 10^{-5}\ \mathrm{m}^2$

| Variable | Description | Value / Formula | Unit |
|---|---|---|---|
| p0 | Static Ambient Pressure | 100000 | Pa |
| V2i | Inlet Velocity | 2.96 | m/s |
| T2i | Static Ambient Temperature | 308.526 | K |
| rho2i | Static Density | 1.13 | kg/m^3 |
| m_dot_a | Air Mass Flow Rate | 0.0453 | kg/s |
| gamma | Ratio of Specific Heats | 1.4 | - |
| R | Gas Constant (Air) | 287 | J/(kg*K) |
| cp | Specific Heat | 1005 | J/(kg*K) |
| LHV | Lower Heating Value | 43000000 | J/kg |
| eta_b | Combustor Efficiency | 0.65 | - |
| Tt4 | Turbine Inlet Temperature | 1250 | K |
| PR | Compressor Pressure Ratio | 1.22 | - |
| eta_c | Compressor Efficiency | 0.8 | - |
| a2i | Speed of Sound at Inlet | 340 | m/s |
| M2i | Mach Number | 0.008705882 | - |
| Tt3 | Comp. Exit Temp (Actual) | 331.0713372 | K |
| pt3 | Comp. Exit Pressure | 122000 | Pa |
| pt4 | Stagnation Pressure (Comb.Exit) | 122000 | Pa |
| f | Fuel-to-Air Ratio | 0.033041979 | - |
| Te | Nozzle Exit Static Temp | 1180.961598 | K |
| ue | Exit Velocity | 372.5146814 | m/s |
| u0 | Flight Speed (Static Test) | 0 | m/s |
| Spec. Thrust | Specific Thrust | 384.8233036 | N/(kg/s) |
| F | Total Thrust | 17.43249565 | N |
| m_dot_f | Fuel Mass Flow Rate | 0.001496802 | kg/s |
| Isp | Specific Impulse | 26207.65159 | s |

| | | | |
|---|---|---|---|
| Pcr * | Pressure for Choked flow | 0.528281788 | NOT CHOKED |
| choke_flag | choke_flag | UNCHOKED | - |
| $M_e$ | Required exit Mach number | 0.540645359 | - |
| $\frac{A_e}{A^*}$ | Area–Mach relation — exit area ratio | 1.269306472 | - |
| | Sonic area (critical area) -- Area_t when throat is sonic | 5.52813E-05 | m^2 |
| $A_e$ | Required nozzle exit area | 7.01689E-05 | m^2 |
| $A_e$ alt | From mass continuity | 7.01689E-05 | m^2 |

Figure 15 State variables estimation at different Stages.

# 2) EFFICIENCY CALCULATION FOR ELECTRIC MICRO JET ENGINE:

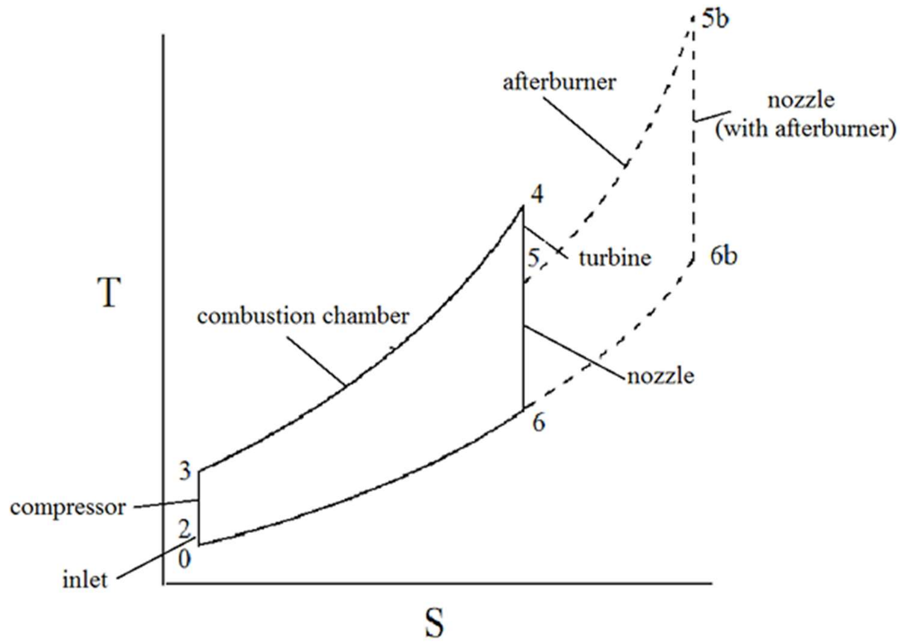



Figure 4: TS diagram of the ideal Brayton Cycle for Turbojet

## Conventional Turbojet Thermal Efficiency

The performance of a standard gas turbine (turbojet) engine is typically analyzed using the Brayton cycle. The thermal efficiency ($\eta_{th}$) defines how effectively the heat energy from the fuel is converted into useful net work.

For an ideal Brayton cycle:

$\eta th = \mathrm{W}net.\ Qin$

where:

- Net Work:

$Wnet = (h3 - h4) - (h2 - h1)$

(the turbine work output minus the compressor work input)

- Heat Input:

$Qin = h3 - h2$

(the enthalpy rise across the combustor)

Here, $h1, h2, h3, h4$ denote the specific enthalpies at the compressor inlet, compressor outlet, turbine inlet, and turbine outlet, respectively.

### The Hybrid Electric-Combustion Concept

In the proposed engine, the turbine is eliminated and the compressor is instead driven by an **electric motor**. This changes the energy balance:

- Compressor work requirement is supplied electrically, not by turbine shaft power.
- The high-enthalpy gas leaving the combustor expands completely through the nozzle, maximizing jet thrust.

Thus, the useful output is **jet kinetic energy → thrust power**, while the inputs are **fuel heat release + electrical motor power**.

### Thrust Power as Useful Work

For a jet engine, the useful work output is the **propulsive power** it delivers to the aircraft. This is known as thrust power:

$Pthrust$=$T$·$Va$

where:

- $T$ = net thrust,
- $Va$ = flight velocity of the aircraft.

This represents the rate of useful work done by the engine to propel the aircraft forward.

## 4. Efficiency Formulations for the Hybrid Engine

### Thermal Efficiency

Thermal efficiency considers only the chemical (fuel) energy input and measures how effectively combustion energy is converted into thrust power.

$$\eta_{th} = \frac{P_{thrust} - Wc}{\dot{Q}_{in}}$$

Substituting:

$$\eta_{th} = \frac{T \cdot V_a - Wc}{\dot{m}_f\, Q_{HV}}$$

where:

- $\dot{m}_f$ = fuel mass flow rate.
- $Q_{HV}$ = heating value of the fuel (kJ/kg).
- $Wc = Work\ done\ by\ turbine$

P thrust = h3-h4, h3 contains all the energy imparted by work from compressor and heat input from addition of fuel, h4 is the heat lost to the surrounding, h3 -h4 is the work done in iso entropic expansion, h3-h4-Wc contains only the fraction of work extracted from heat , so the ratio (h3-h4-Wc)/Q is the fraction of heat converted to work , (Thermal Efficiency )

## Overall Energy Efficiency

The **overall efficiency** accounts for both chemical and electrical energy inputs:

$\eta overall = PthrustQin + Pmotor$

Substituting:

$$\eta_{overall} = \frac{T \cdot V_a}{\dot{m}_f\, Q_{HV} + P_{motor}}$$

where:

- $P_{motor}$ = electrical power supplied to drive the compressor.

## Note on Motor Power

The instantaneous **rotational kinetic energy** of the compressor is:

$$E_{rotor} = 1/2\, I\omega^2$$

However, for steady-state operation, the relevant input is **power**, not stored energy. The motor must continuously supply the compression work:

$$P_{motor} = \dot{m}_{air}\, (h_2 - h_1)$$

where:

- $\dot{m}_{air}$ = air mass flow rate through the compressor,
- $(h_2 - h_1)$ = enthalpy rise across the compressor.

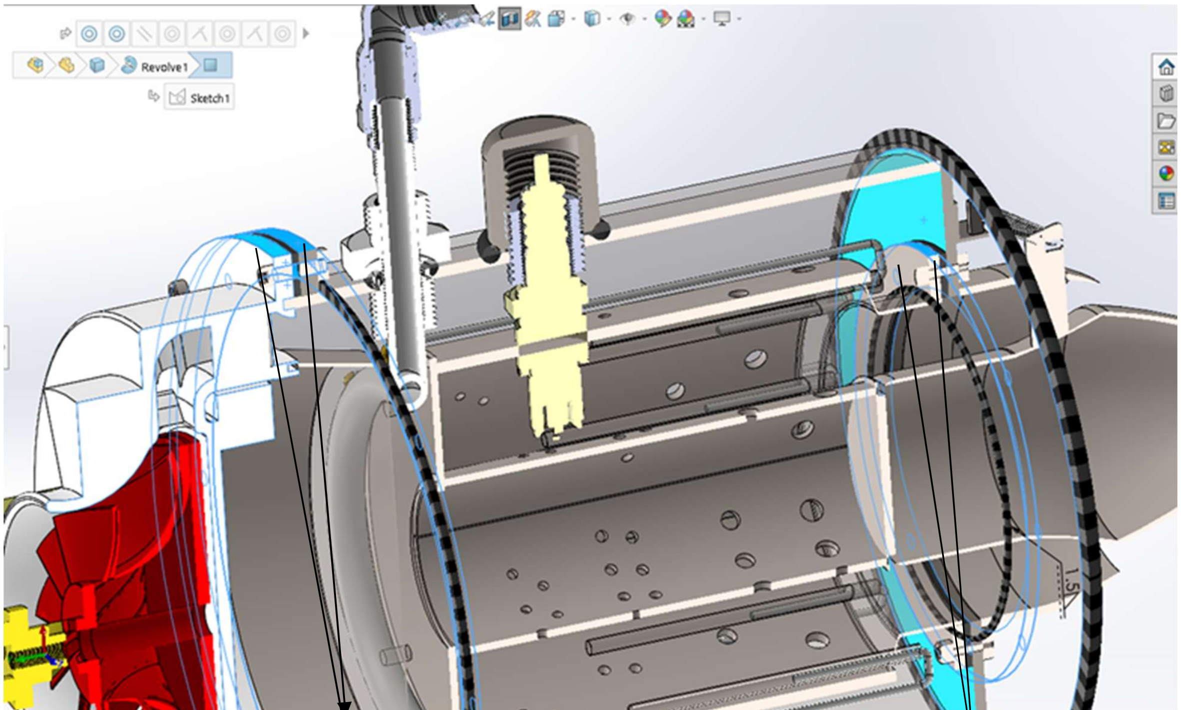
## Summary of Efficiency Equations

- **Thermal Efficiency** (fuel only):

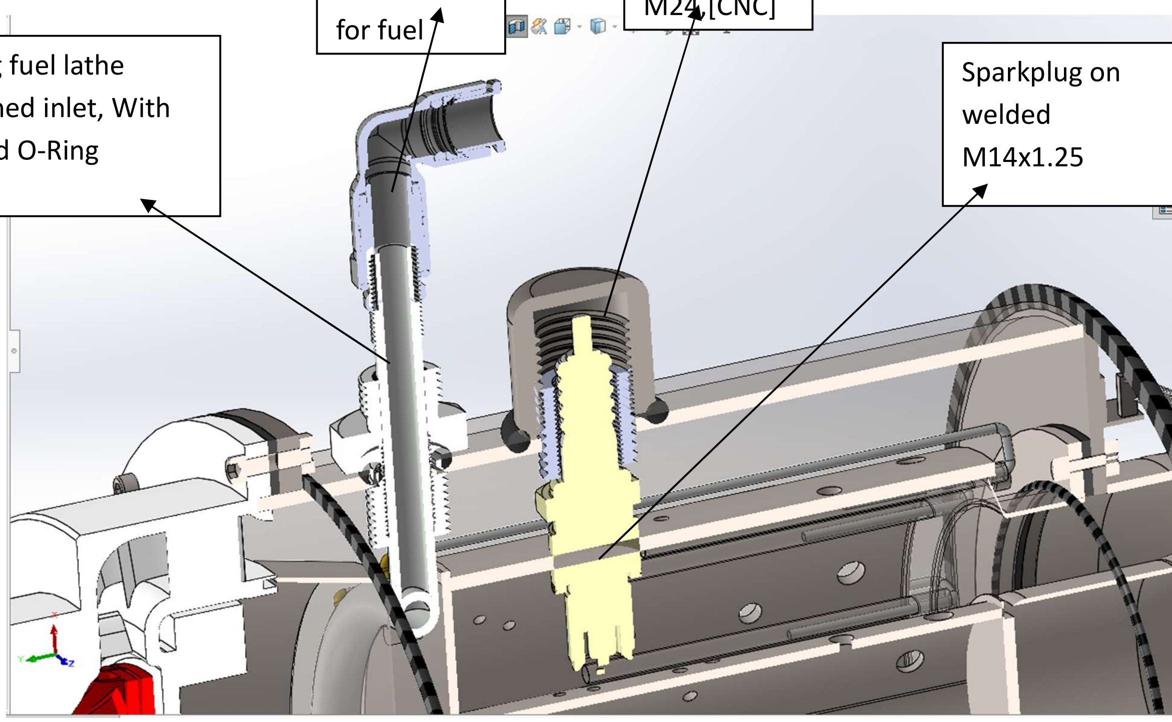
$$\eta_{th} = \frac{T \cdot V_a - Wc}{\dot{m}_f\, Q_{HV}}$$

- **Overall Efficiency** (fuel + motor):
- $\eta_{overall} = \frac{T \cdot V_a}{\dot{m}_f\, Q_{HV} + P_{motor}}$

## 6.7 CAD DESIGN:




Gasket sealed Joints between combustion chamber and compressor(left) and combustion chamber and outer casing(right).



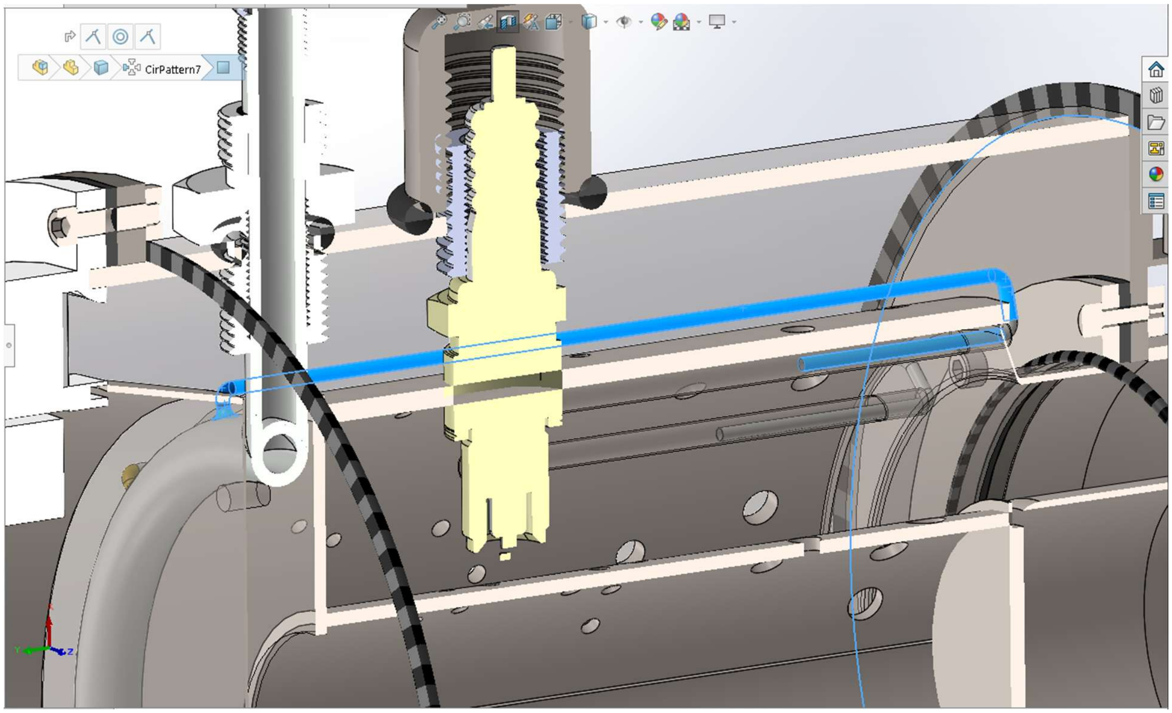



Figure 16 SS 1.2MM ID Injection Tubes (Blue), passing over the outer liner, carries fuel which absorbs heat from the outer liner evaporating the fuel before injection for better mixing.

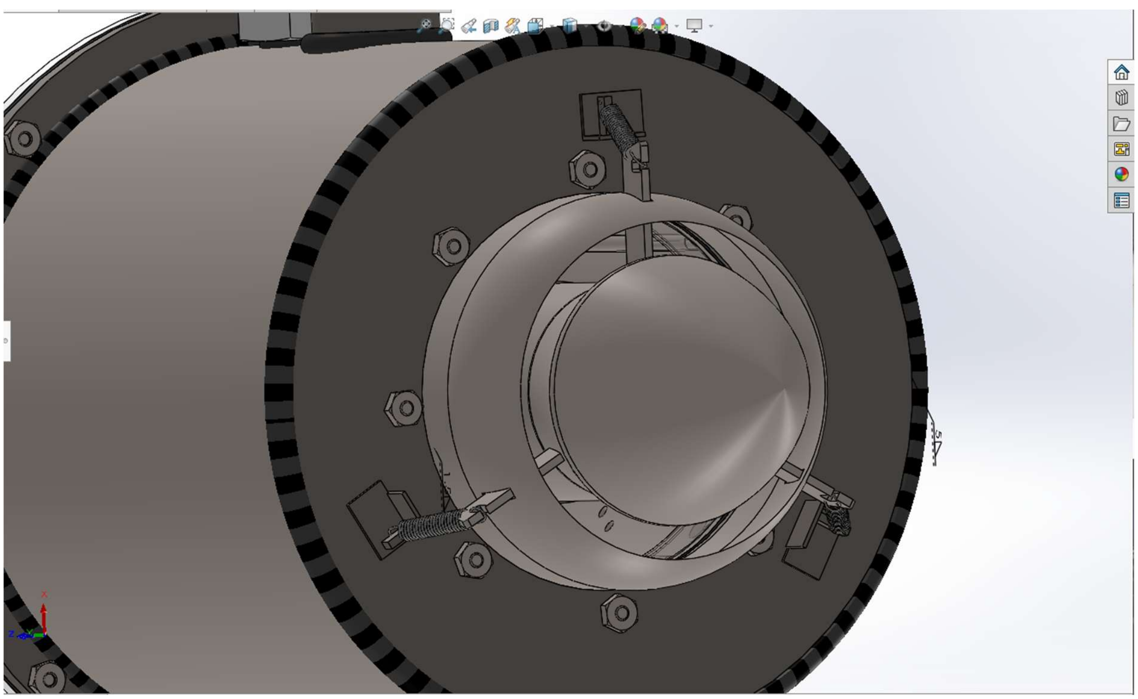


Figure 17 Nozzle assembled to outer casing with springs.

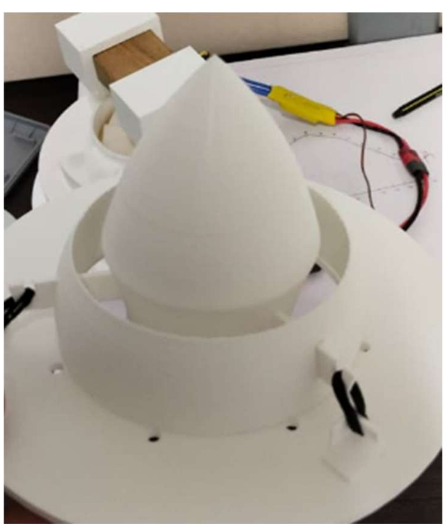
# 7. RESULTS AND DISCUSSION:

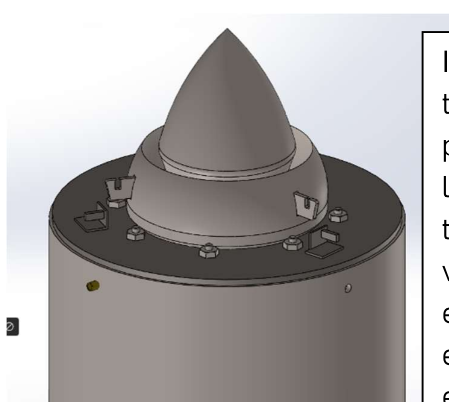
## 7.1 Microjet Engine Fabrication Process – Stepwise Manufacturing Workflow

Figure 18

Acrylic sheets, mesh for test rig enclosure, outer liner (5'), sheet metal (1mm mild steel). Rubber gaskets, drill bits (HSS [2-6 mm]), nut and bolts for assembly.

### 5) Exhaust nozzle





It takes the high-temperature, high-pressure gas exiting the liner and accelerate it to the highest possible velocity as it exits the engine. This high-speed exhaust jet produces the engine's thrust.

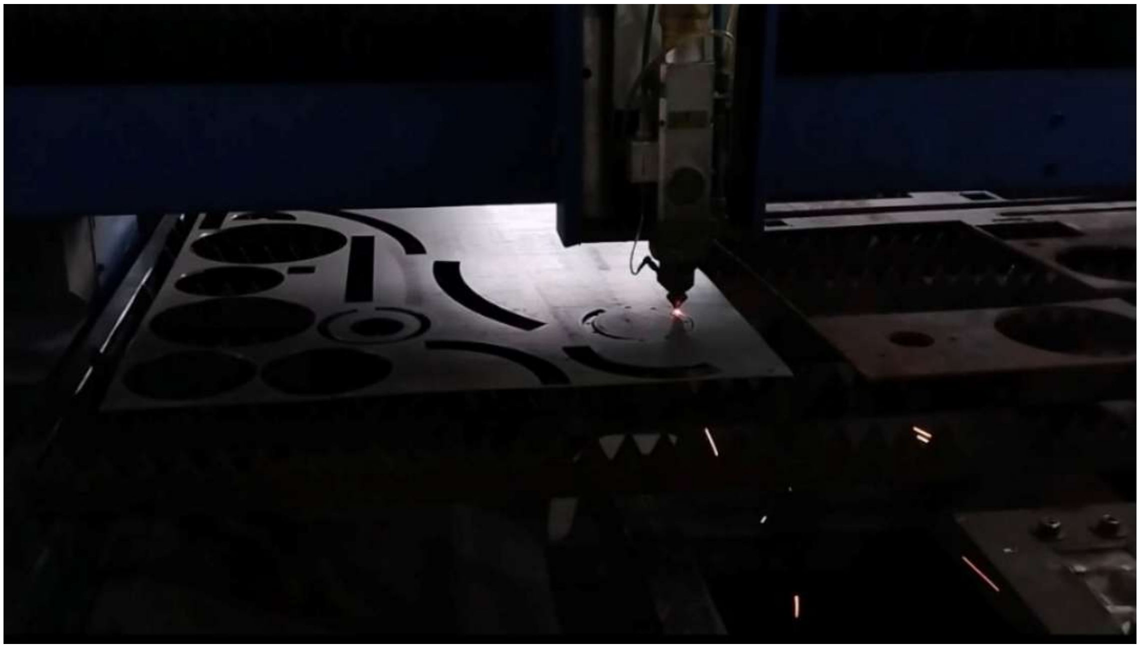


Figure 19 Figure Laser Cutting of Mild Steel Sheet (1 mm Thick)

Figure 19 Preparation and Finishing of Laser-Cut Parts

Figure 18  Outer Liner Marking and Drilling on Mild Steel Tube (15.6 cm Diameter)

Figure 20 Arc Welding of Inner Liner, Outer Liner, and Outer Casing

Figure 21 20 Nutted Joint Assembly Between Combustion Chamber and Outer Casing.

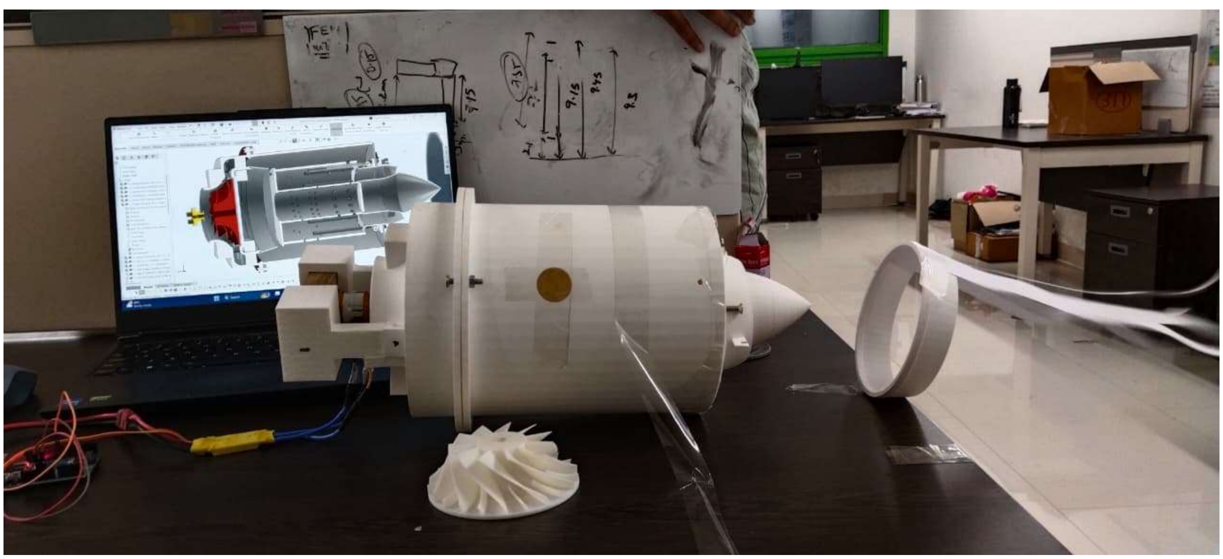


Figure 21 There were some leakages at compressor combustion chamber joints, use of rubber gaskets can help with better sealing and also the compression is not as much as expected and velocity is high so we would improve the diverging duct design and to increase the pressure of moving fluid, in further iterations

Figure_22 Brazing of Evaporator Tubes and Fuel Injection Tubes

Figure 23  Combustion Chamber Assembly with Integrated Fuel Tubes

Figure 23 Plasma Cutting for Spark Plug Port

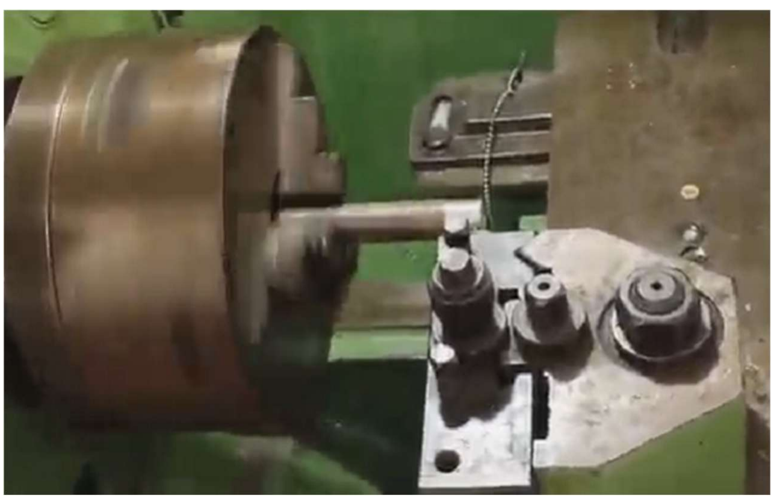
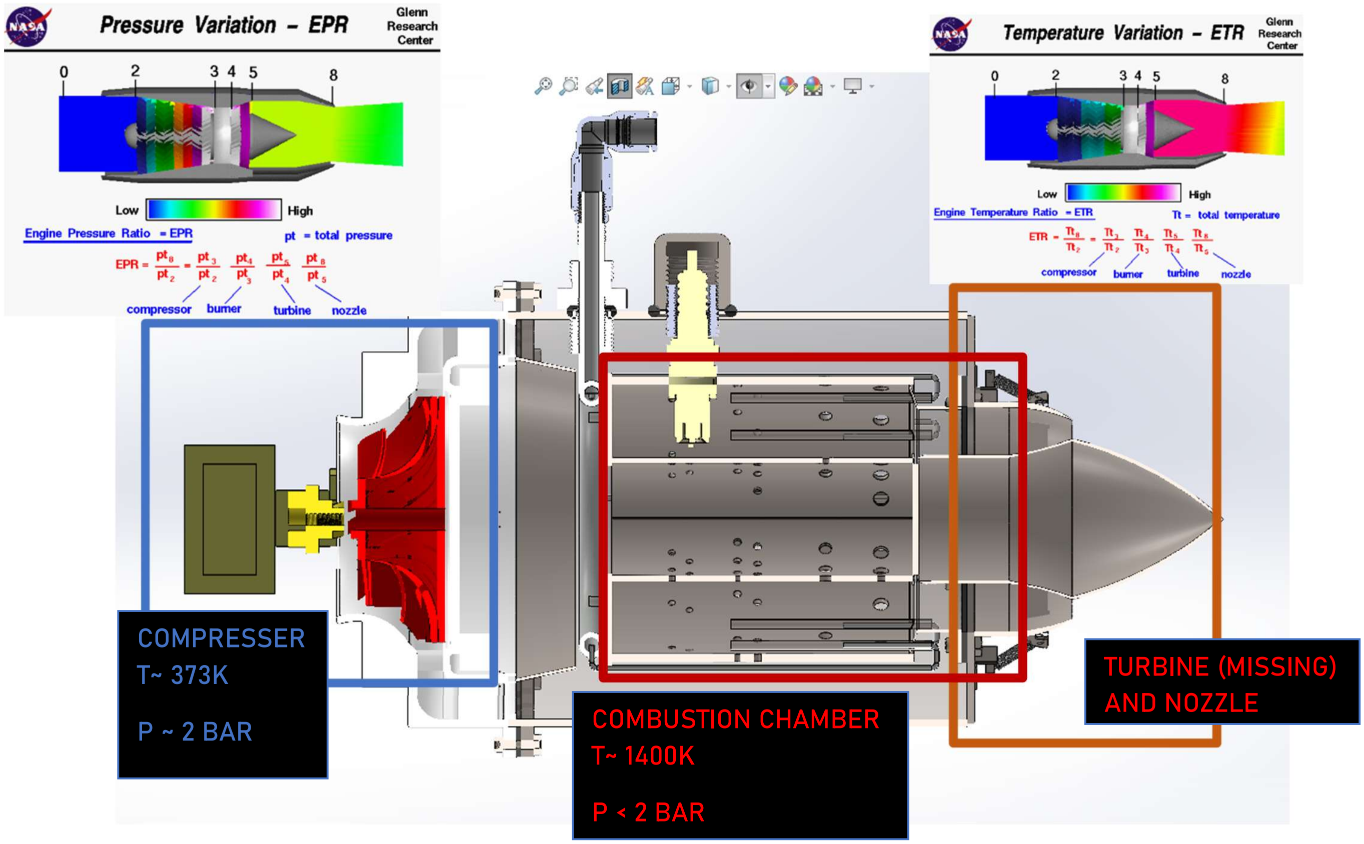


Figure 22  Fuel Inlet Tube Fabrication on Lathe Machine

*Figure 27 Nozzle Forming and Welding Operations*


Figure 24 Protective Enclosure Fabrication for Engine Testing

## 7.2 CONTROL OF MICRO JET ENGINE:




## COMPONENTS TO CONTROL:

Inputs:

1)Compressor motor -BLDC PWM Input

2)Fuel Pump – DC Motor PWM input

3)Spark Plug – Digital (0/1) input

Outputs: (Monitoring State Variables)

1)Temperature Transducer – Analog Input

2)Pressure Transducer - Analog input

Figure 27  SPARK PLUG AND FUEL

Figure 28 FUEL INJECTION SYSTEM SPARK PLUG TRIGGERED USING RELAY.

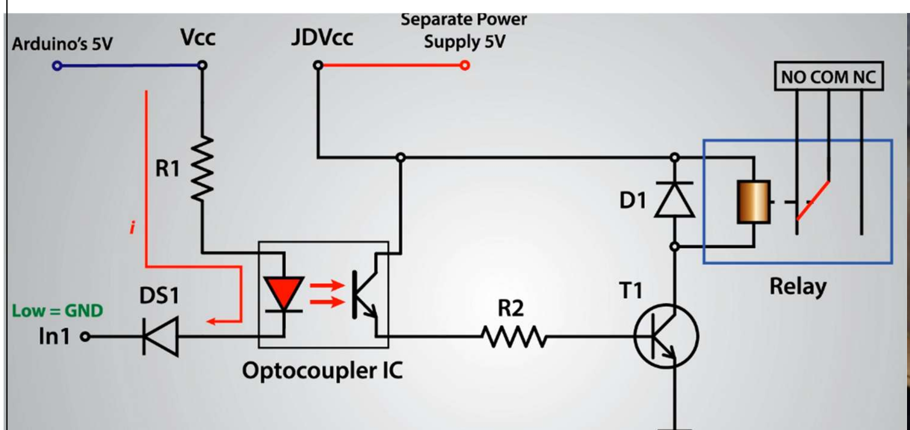
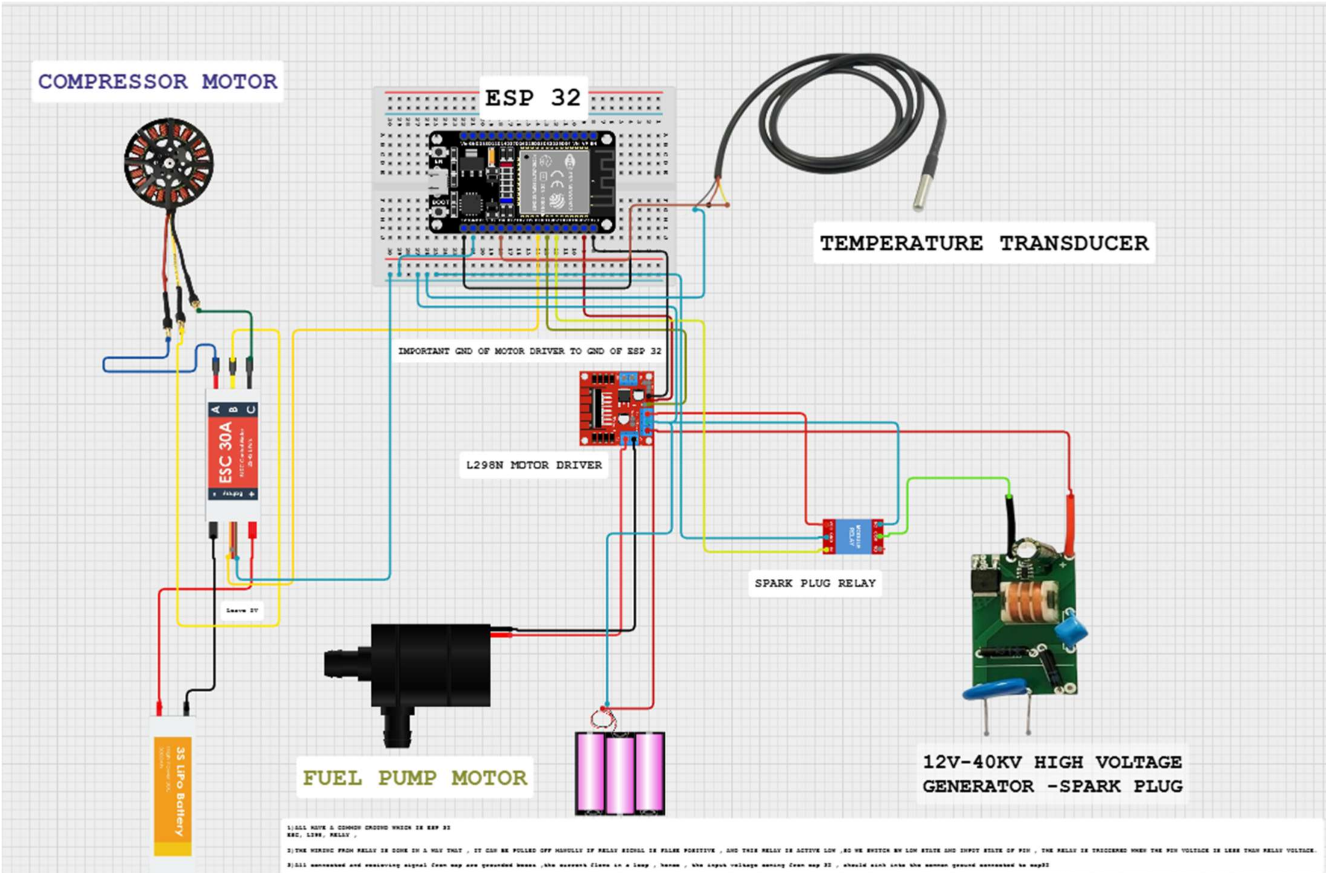




Figure 25 WORKING OF RELAY AND SENSOR: TEMPERATURE THERMOCOUPLE- (J type)

## CONTROL CIRCUIT:



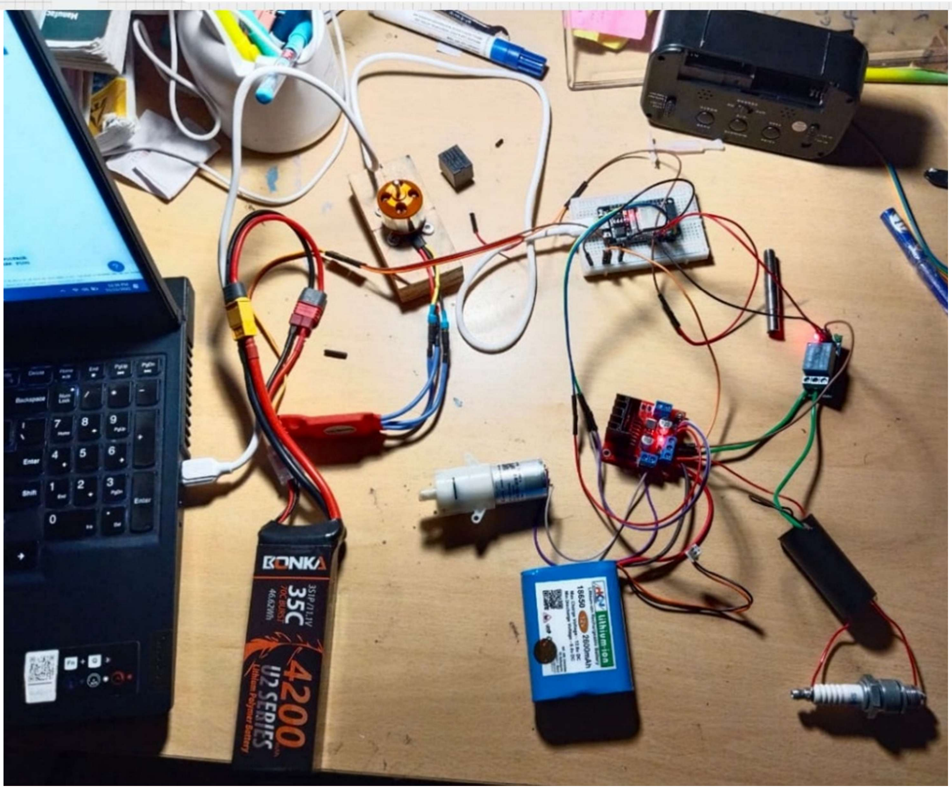
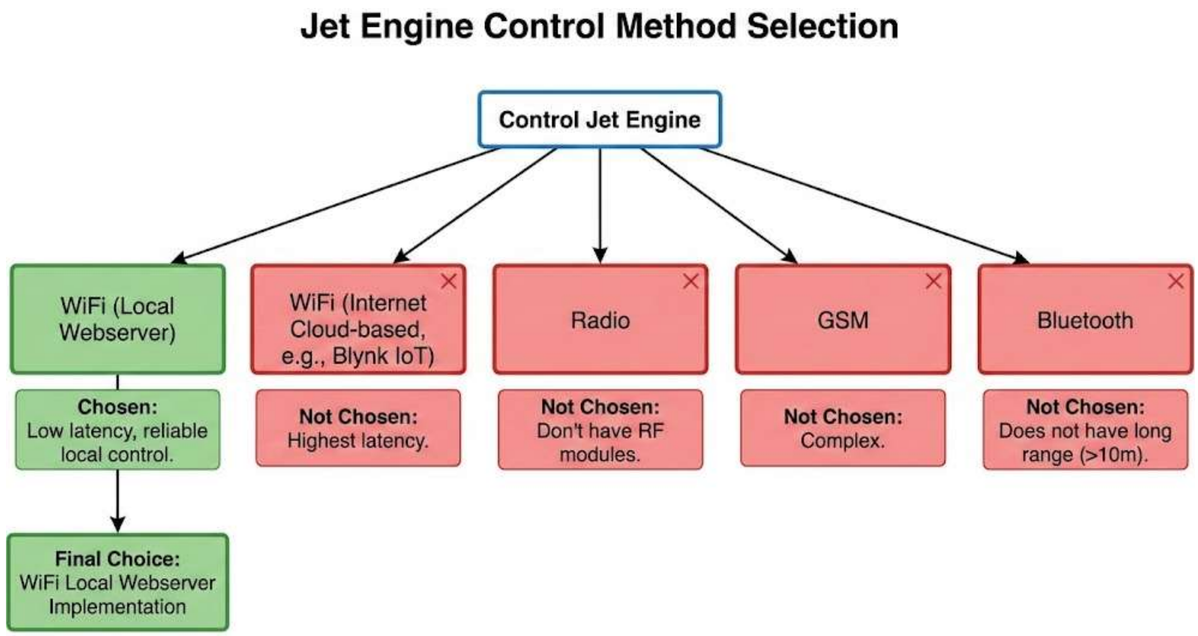
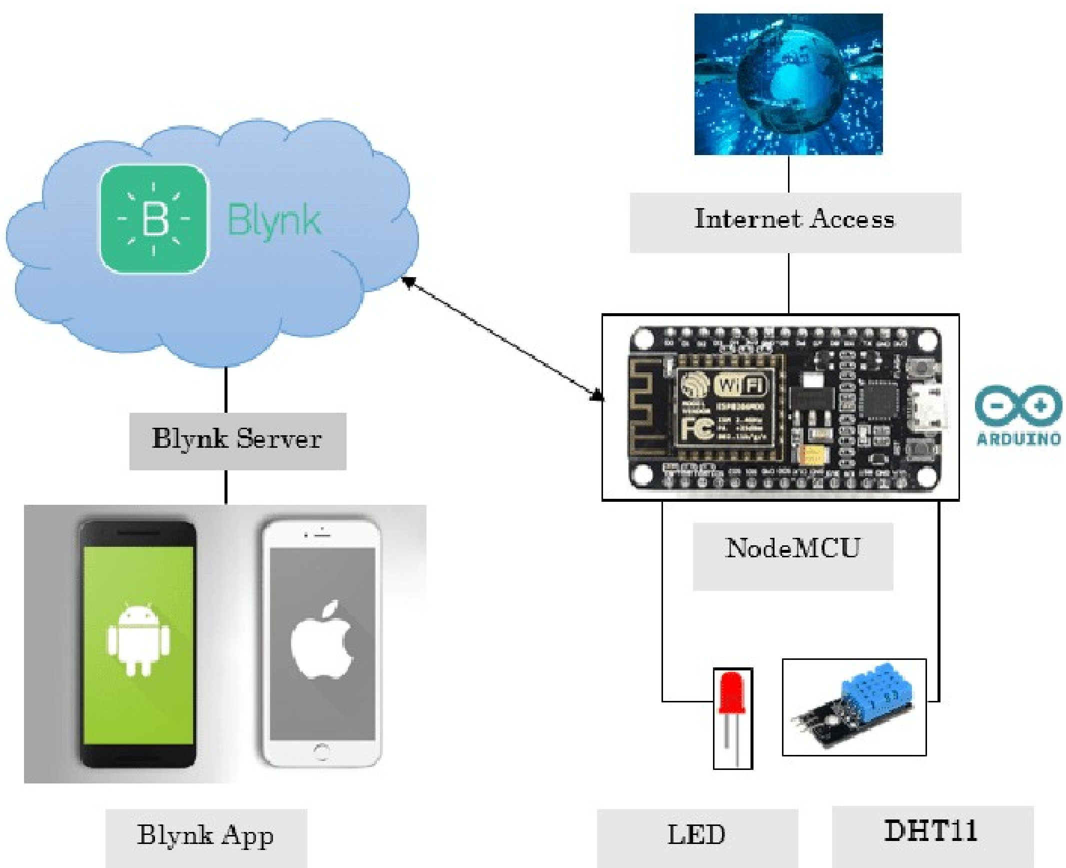




Fig 30 Circuit to control compressor motor, fuel pump and spark plug


Jet Engine Control Method Selection
Control Jet Engine
WiFi (Local Webserver)
WiFi (Internet Cloud-based, e.g., Blynk IoT)
Radio
GSM
Bluetooth
Chosen: Low latency, reliable local control.
Not Chosen: Highest latency.
Not Chosen: Don't have RF modules.
Not Chosen: Complex.
Not Chosen: Does not have long range (>10m).
Final Choice: WiFi Local Webserver Implementation


Blynk
Blynk Server
Blynk App
Internet Access
NodeMCU
ARDUINO
LED
DHT11

# ESP32 Engine Controller - Quick Guide

(BETA)

## Hardware Connections

| Component | ESP32 Pin | Notes |
|---|---|---|
| **ESC Signal** | GPIO 18 | Yellow wire from ESC |
| **L298N EN** | GPIO 23 | Speed control (PWM) |
| **L298N IN1** | GPIO 22 | Direction pin 1 |
| **L298N IN2** | GPIO 21 | Direction pin 2 |
| **Spark Relay** | GPIO 26 | Controls relay to spark plug |

### L298N Motor Driver Wiring:

- **12V Power** → L298N 12V input
- **GND** → Common ground (ESP32 + L298N + 12V supply)
- **Motor** → L298N OUT1 & OUT2
- **EN, IN1, IN2** → ESP32 as shown above

---

## Blynk App Setup

| Virtual Pin | Widget | Purpose |
|---|---|---|
| **V1** | Slider (0-100) | Manual ESC control |
| **V2** | Slider (0-100) | Manual fuel pump |
| **V3** | Button | Manual spark on/off |
| **V4** | LED | Manual mode indicator |
| **V5** | LED | Auto mode indicator |
| **V10** | Slider (0-100) | Auto mode ESC target |
| **V11** | Slider (200-60000) | Auto spark period (milliseconds) |
| **V12** | Slider (0-100) | Auto mode pump target |
| **V13** | Button | Switch Manual ↔ Auto |

## How It Works

### **Manual Mode** (Default)

You're the boss. Control everything yourself:

- Slide V1 to control motor speed (0-100%)
- Slide V2 to control fuel pump (0-100%)
- Press V3 to turn spark plug on/off
- For testing on manual operation

### **Auto Mode** (Press V13 to activate)

The system runs a pre-programmed startup sequence automatically:

1. **Gentle Startup**: Ramps ESC through 10% → 20% → 30% → 50%
2. **Noise Check**: Holds at 50% for 10 seconds (listen for issues)
3. **Full Power**: Increases to your target ESC% (default 80%)
4. **Stabilize**: Waits 10 seconds at target speed
5. **Ignition**: Starts spark plug cycling (default every 2 seconds)
6. **Fuel Injection**: Starts fuel pump at target% (default 50%)
7. **Running**: System keeps running until you switch back to manual

**Before starting Auto Mode**, set your targets:

- V10: How fast you want the motor (default 80%)
- V11: How often spark fires (2000ms = spark toggles every 1 second)
- V12: How much fuel to inject (default 50%)

---

## Quick Start

1. **Upload code** with your WiFi credentials and Blynk token
2. **Wire everything** according to the table above
3. **Power up** - System starts in Manual Mode
4. **Test manually** - Try sliders V1, V2, and button V3
5. **Set auto parameters** - Adjust V10, V11, V12 if needed
6. **Go auto** - Press V13 to start automatic sequence
7. **Emergency stop** - Press V13 again to return to Manual (everything stops)

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Codes :

MicroJetEngineDevelopmentEngineeringProject/Blynk Control code/ESP 32-Blynk_Auto & Manual_control_(Beta).cpp at main · BillKerman/MicroJetEngineDevelopmentEngineeringProject

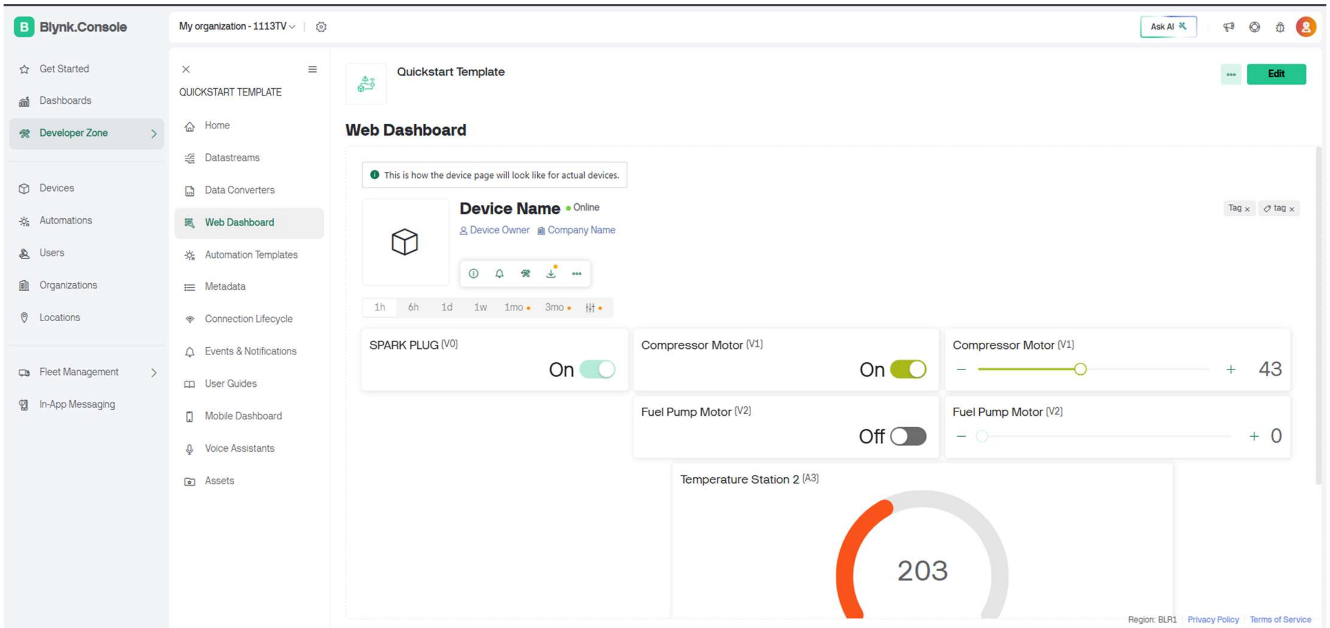


Figure 27 WEB DASHBOARD TO CONTROL OVER INTERNET

## II)Hosting a local web server:

## Working of Local Web server:

### 1. WiFi Connection

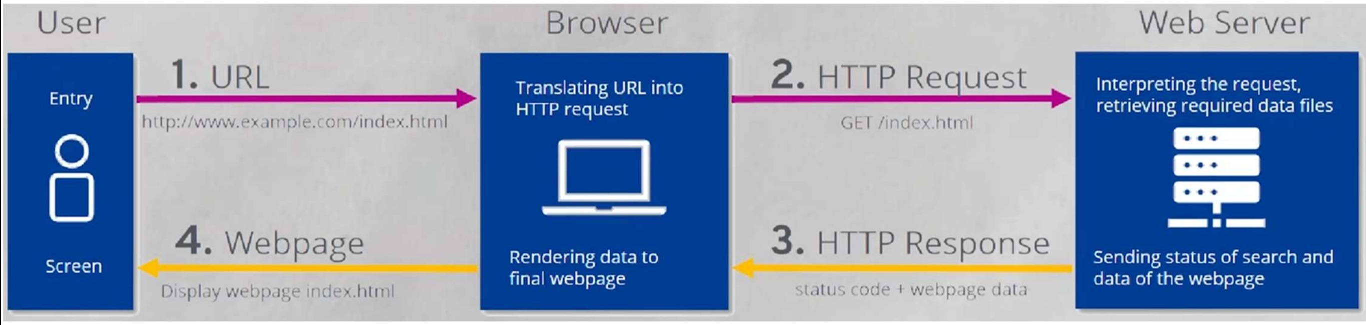
- Function: WiFi.begin(ssid, password)
- Description: This initializes the WiFi module and attempts to handshake with the router using the provided network credentials to establish a link.

### 2. IP Assignment (DHCP)

- Function: WiFi.localIP() (Used to verify assignment)

Figure 32 28 WEB DASHBOARD TO CONTROL OVER INTERNET

- Description: The router's DHCP server automatically leases a unique IP address to the board, which this function retrieves to confirm a successful network **connection.**




### 3. mDNS (Multicast DNS)

- Function: MDNS.begin("host_name")
- Description: This starts the multicast DNS responder, linking a custom name (like rover.local) to the dynamic IP so the address never needs to be hardcoded.

### 4. Server Initialization & HTML

- Function: server.on("/", handle_function)
- Description: This links a specific URL path (like the root /) to a specific function containing your HTML code, telling the server what to prepare when accessed.

### 5. Client Request & Connection

- Function: server.handleClient()
- Description: Placed in the main loop, this function actively checks for incoming TCP requests from a browser and delegates them to the correct handling function.

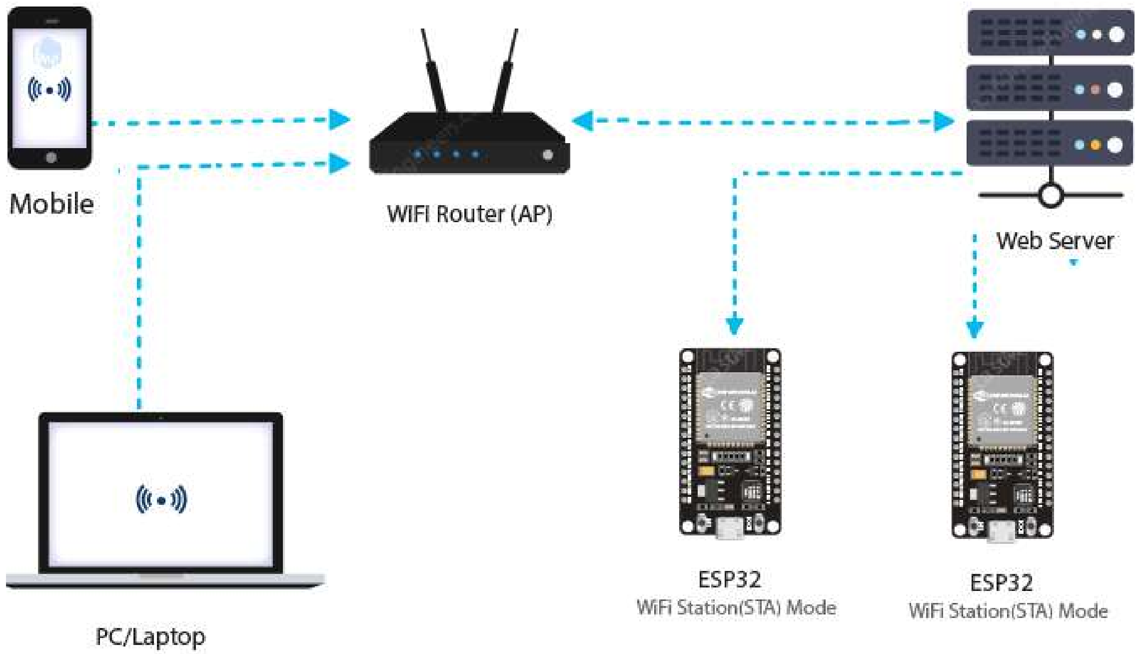
### 6. Response & Rendering

- Function: server.send(200, "text/html", html_code)
- Description: This constructs the HTTP response packet containing a status code (200 OK) and the HTML body, transmitting it back for the browser to display.

### 7. Station Mode (STA)

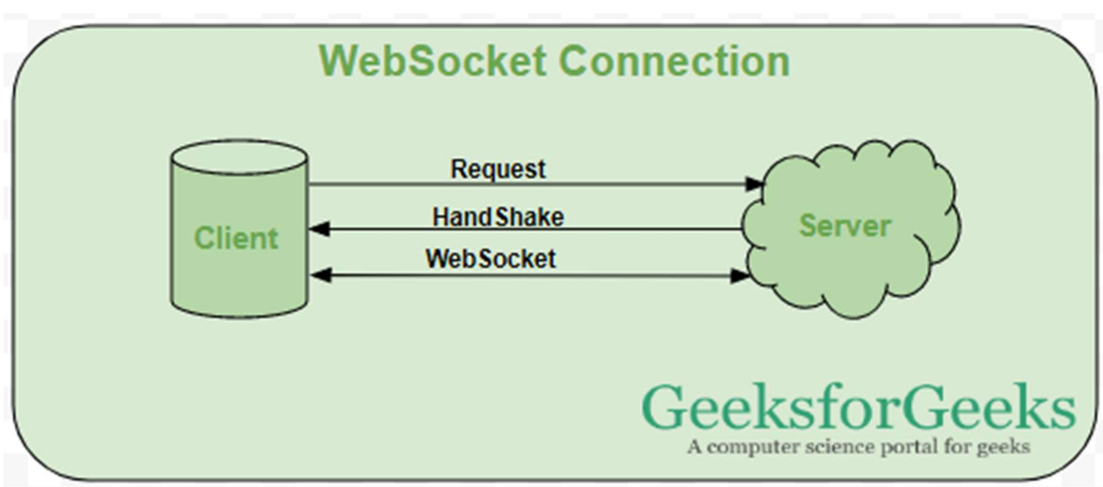
- Function: WiFi.mode(WIFI_STA);

- **Description:** This mode configures the ESP32 to act as a client device that connects to an existing WiFi router, allowing it to access the internet or the local network.




### 8. Soft Access Point Mode (SoftAP)

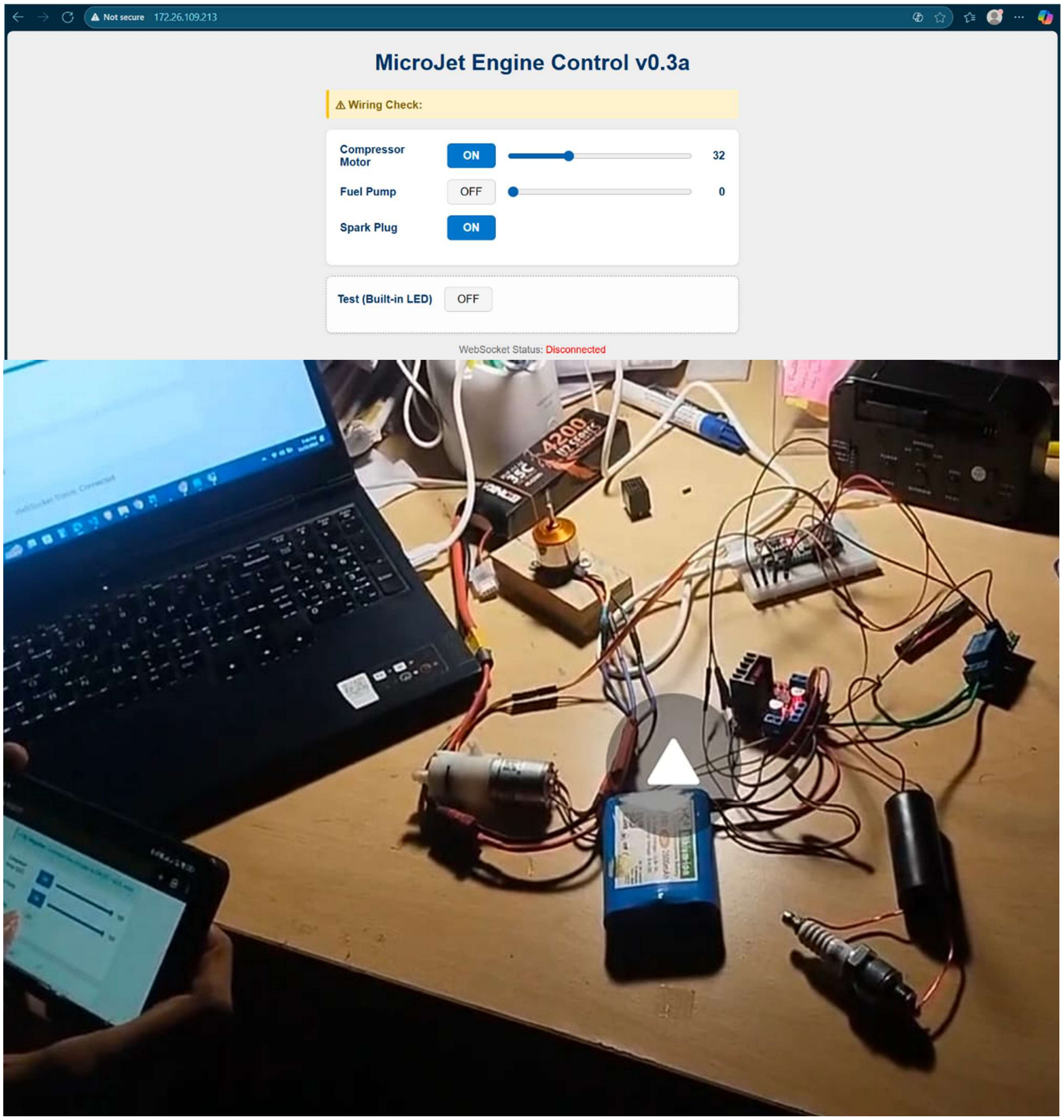
- **Function:** WiFi.softAP(ssid, password);
- **Description:** This mode turns the ESP32 into a hotspot (like a router) with its own network name (SSID), allowing other devices to connect directly to it without needing an external router.



# Workflow of Data moving from User (Website) to Client (ESP 32) and vice versa:

## A custom web dashboard to control MJE.

Integrating WebSocket:




## Core Concepts

1. **WebSocket (Two-Way Communication)** Unlike standard HTTP where the client must ask for every update, a WebSocket creates a permanent, open channel (full-duplex).

This allows the server to push data to the client instantly without the page ever needing to reload.

**2. DHCP (Dynamic Host Configuration Protocol)** DHCP is the network "manager" that automatically assigns a unique IP address to your ESP32 and connecting clients. It ensures everyone has a valid address so they can find each other to establish the WebSocket connection.

**3. Broadcasting** This feature allows the server to send a single message (like a sensor reading) to *every* connected client simultaneously, ensuring all users see the exact same data at the same time.

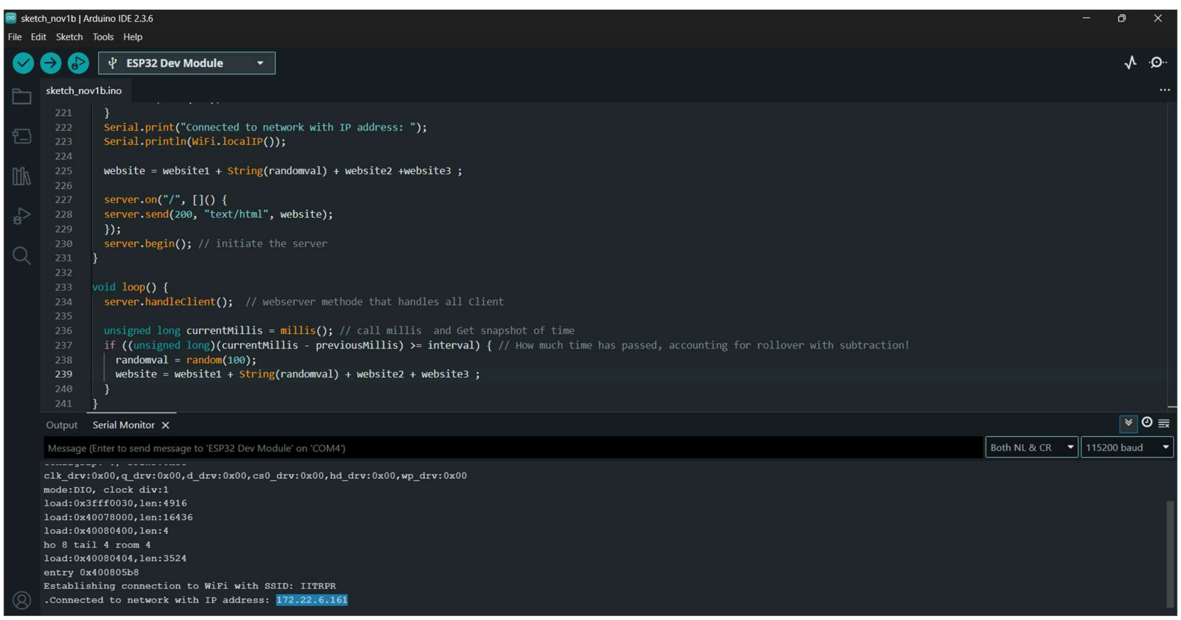
## Working Flow & Functions

### 4. Client Connects to Web Server

- **Action:** The browser requests the webpage using the ESP32's IP address.
- **Function:** server.on("/", ...) serves the HTML file which contains the JavaScript code needed to start the WebSocket.

### 5. Webpage Connects to WebSocket

- **Action:** The JavaScript on the page immediately upgrades the HTTP connection to a WebSocket connection.
- **Function (JS):** new WebSocket ('ws://' + window.location.hostname + '/ws') initiates the handshake to open the permanent line.



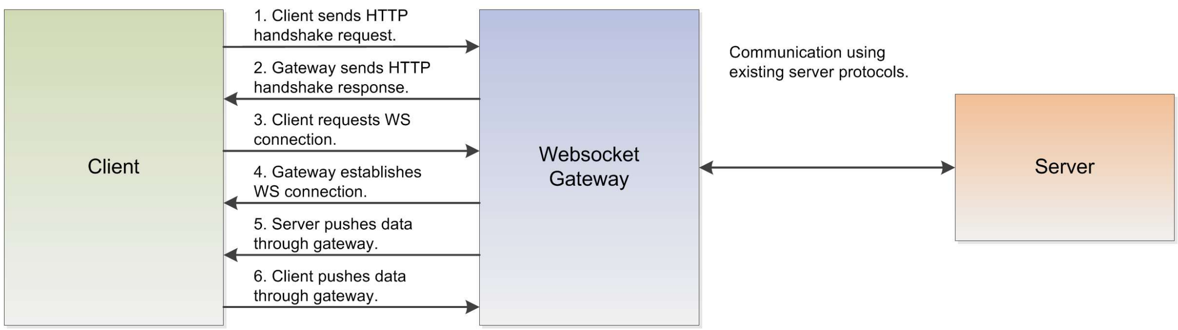

### 6. Client Event Listeners

- **Action:** The webpage waits for user actions (like pressing a button) or incoming messages from the server.

- **Function (JS):** document.getElementById('btn').addEventListener('click', ...) detects clicks to trigger data sending.

7. Sending Variables via JSON

- **Action:** When an event triggers, the client packages data into a JSON string (e.g., {"speed": 100}) and sends it.
- **Function (JS):** websocket.send(JSON.stringify(data)) converts the data object into text and transmits it.




8. WebSocket Loop

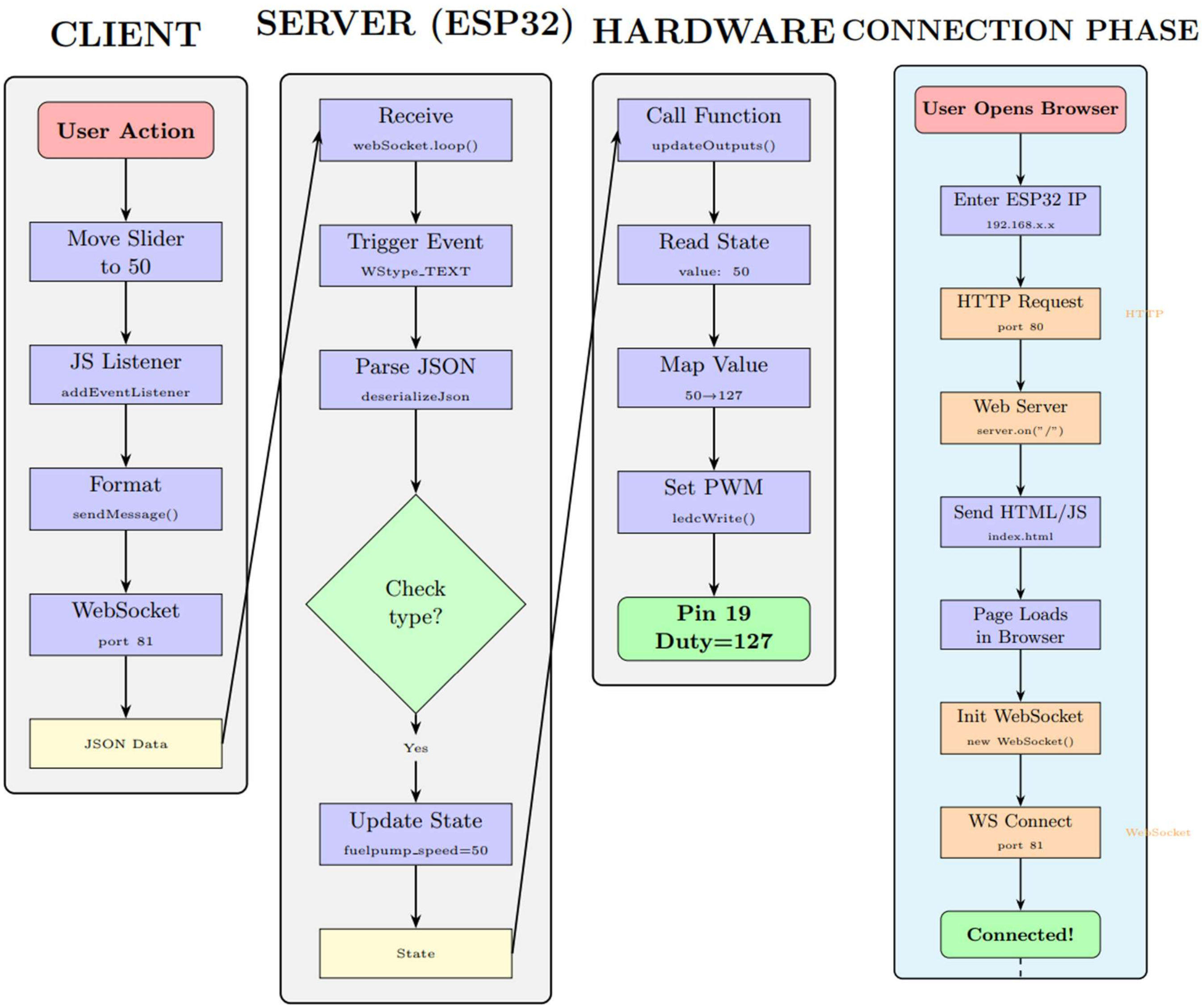
- **Action:** The ESP32 must constantly keep the connection alive and check for any new data packets arriving from clients.
- **Function:** webSocket.loop() is placed inside void loop () to actively manage traffic and maintain the heartbeat.

9. WebSocket Event & Parsing

- **Action:** When data arrives, this callback function triggers, detects the message type (text), and decodes the JSON.
- **Function:** webSocketEvent(type, payload, length) receives the raw data; JSON.parse(payload) (via ArduinoJson) extracts the values.

10. Assign Variables & Update Outputs

- **Action:** The code takes the parsed value (e.g., speed) and applies it to the hardware (motors/LEDs).
- **Function:** analogWrite(pin, parsedValue) updates the physical output based on the instruction received from the web page.


CLIENT
SERVER (ESP32)
HARDWARE
CONNECTION PHASE
User Action
Move Slider to 50
JS Listener
addEventListener
Format
sendMessage()
WebSocket
port 81
JSON Data
Receive
webSocket.loop()
Trigger Event
WStype_TEXT
Parse JSON
deserializeJson
Check type?
Yes
Update State
fuelpump_speed=50
State
Call Function
updateOutputs()
Read State
value: 50
Map Value
50→127
Set PWM
ledcWrite()
Pin 19 Duty=127
User Opens Browser
Enter ESP32 IP
192.168.x.x
HTTP Request
port 80
HTTP
Web Server
server.on("/")
Send HTML/JS
index.html
Page Loads in Browser
Init WebSocket
new WebSocket()
WS Connect
port 81
WebSocket
Connected!

# 8. Finding and Conclusions:

## 8.1 Limitations:

- Absence of a proper shaft–turbine coupling and insufficient motor-holder stiffness caused vibration, misalignment, and dynamic instability, leading to impeller–duct contact, increased tip clearance, and leakage. These mechanical issues limited achievable RPM, reduced compressor loading, and restricted the attainable pressure ratio and stagnation pressure.
- Low RPM combined with leakage resulted in a lower compressor pressure ratio and mass flow rate, directly producing much lower thrust than predicted and limiting the overall engine operating window.
- The combustion chamber material's low melting/softening temperature restricted safe thermal operation; high-temperature exposure risked deformation, forcing reduced fuel flow and shorter test durations.
- Dimensional variability in laser-cut parts, difficult welding of thin injection tubes (burn-through during arc welding), and weak, poor-finish arc-brazed joints led to improper fitting, reduced sealing quality, lower aerodynamic smoothness, and lower structural durability.
- Multiple electronic components (ESP32, ESC, Arduino, BLDC motor, voltage generator) failed due to inadequate electrical protection, transient spikes, and poor grounding, reducing system reliability during tests.
- Difficulty sourcing required pipes, tubes, and high-temperature materials restricted design choices, while outsourcing several fabrication steps introduced delays and reduced precision in critical components.
- Collectively, these mechanical, thermal, fabrication, and electronic limitations resulted in lower RPM, reduced pressure ratio, lower mass flow, significantly lower thrust, and reduced overall system reliability.

## 8.2 Future Progress:

- Conduct static fire tests to measure state variables (pressure, temperature, density, velocity) across all stages of the engine and compare them with theoretical predictions to quantify losses and deviations.
- Use the collected experimental data for iterative design refinement, targeting improved mass flow rate, higher compressor pressure ratio, reduced leakage, and closer alignment with theoretical performance models.
- Redesign the compressor–rotor assembly by introducing a properly supported shaft, precision bearings, and a rigid mounting base to eliminate vibration, minimize radial runout, and maintain controlled tip clearances.

- Improve the compressor blade and diffuser geometry to enhance aerodynamic loading, increase pressure rise, and stabilize flow under higher RPM conditions.
- Integrate a complete sensor suite (RPM, pressure transducers, thermocouples, thrust load cell, and possible flow measurement) to enable high-resolution data acquisition for performance analysis and data analytics.
- Explore a secondary gaseous fuel inlet (e.g., butane or propane) for smoother engine start-up and stable combustion initiation before switching to liquid fuel.
- Develop and fabricate a turbine stage that can extract energy from exhaust flow to drive the compressor, increasing achievable RPM and enabling compressor re-start capabilities in a turbojet configuration.
- Investigate electro-turbojet / electro-turbo hybrid systems, where electrical power assists the compressor during start-up, transient operation, or low-speed regimes to improve overall responsiveness and performance.

# 9. References:


[1] A. H. Lefebvre and D. R. Ballal, *Gas Turbine Combustion: Alternative Fuels and Emissions*, 3rd ed. Boca Raton, FL: CRC Press, 2010.

[2] H. I. H. Saravanamuttoo, G. F. C. Rogers, and H. Cohen, *Gas Turbine Theory*, 6th ed. Harlow, UK: Pearson Education, 2008.

[3] B. C. Meyers, "The preliminary design of an annular combustor for a mini gas turbine," in *Proceedings of the XXII International Society for Air Breathing Engines (ISABE) Conference*, Phoenix, AZ, USA, Oct. 2015, Paper ISABE-2015-20032.

[4] C. A. Putra, "Design and analysis of annular combustion chamber for a micro turbojet engine," Course Report, AE 5014 Aerodinamika Propulsi, Bandung Institute of Technology, Indonesia, Apr. 2020.

[5] R. S. E. Mohammed, "Design and analysis of annular combustion chamber for a micro turbojet engine," *International Journal of Mechanical, Industrial and Aerospace Sciences*, vol. 13, no. 4, 2019.

[6] R. R. Santoso, V. Wuwung, R. F. N. Annisa, and R. S. Kartanegara, "Design of mini turbojet engine combustion chamber liner with 200N static thrust," *Indonesian Journal of Aerospace*, vol. 21, no. 2, pp. 65–82, Dec. 2023. doi: 10.55981/ijoa.2023.38.

# 10. VIEWS OF MICRO JET ENGINE:

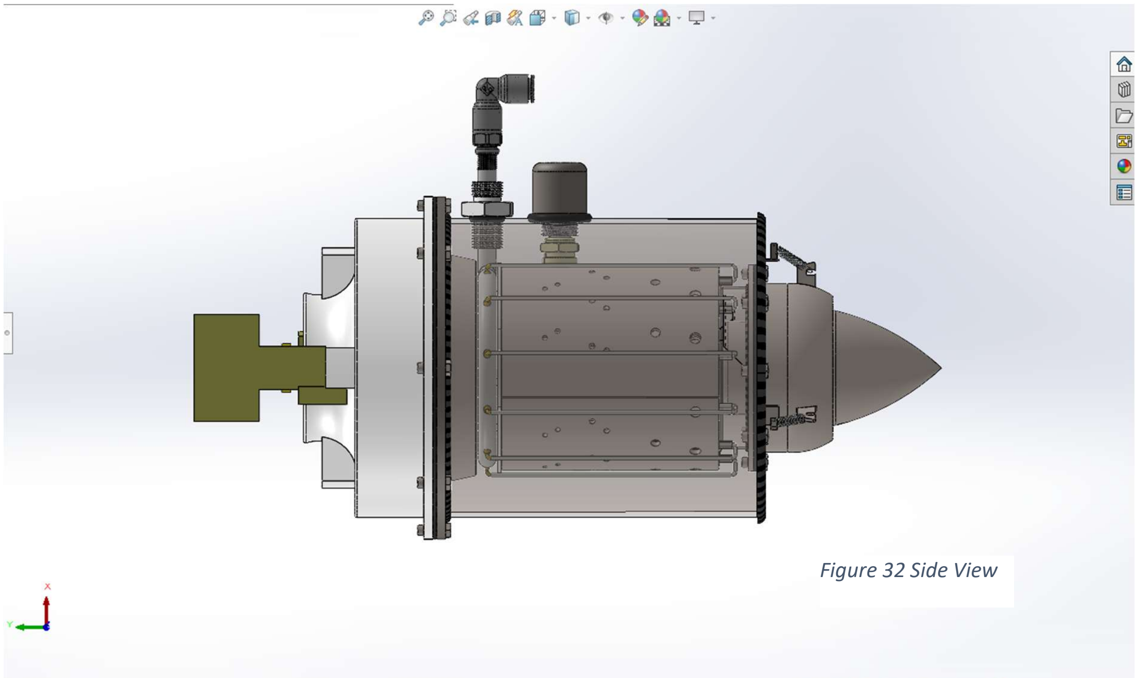


*Figure 32 Side View*

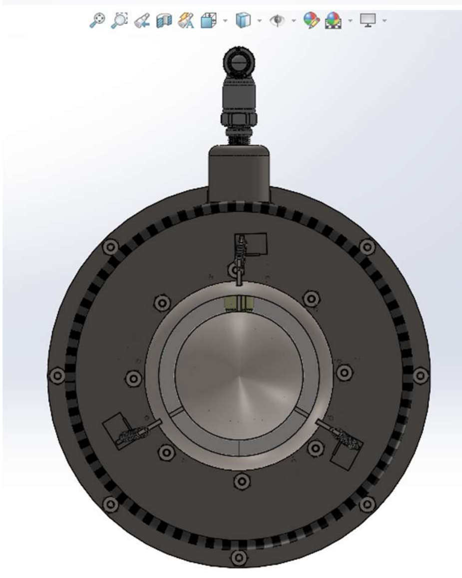


*Figure 31 Rear View*

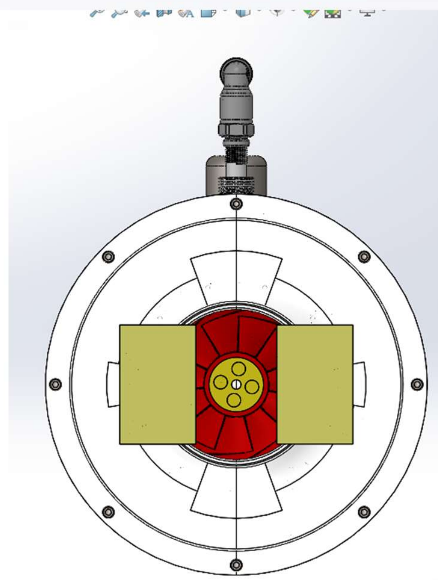


*Figure 30 Front View*

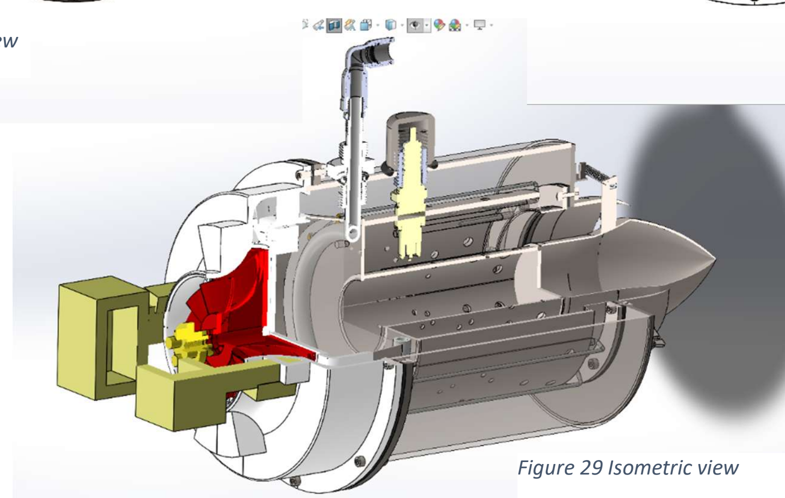


*Figure 29 Isometric view*